
Operational Amplifiers

A Solving Manual

IESE Manual 2 of 3

Typeset in L^AT_EX with circuitikz and pgfplots.

How to use this manual

Typical treatments present dozens of numbered examples but leave the *method* implicit. This manual makes the method explicit. The body is organised by **topic**; inside each topic, every distinct *type* of problem gets a **Protocol** — a numbered recipe you can re-run on any problem of that type — followed by one fully **Worked Example** (the canonical one) and **Exercises** (the near-duplicates) with solutions. A one-paragraph theory recap sits at the point of use; the full derivation lives in a cross-referenced **appendix**, so the body stays a fast solving reference while the “why” is always one hyperlink away. Any term or symbol that is used before it is explained is collected, with a plain-language definition, in the **Glossary** (Appendix F); it ends with a one-page symbol key.

Key Idea

The two golden rules behind almost every OPAMP circuit with negative feedback: the differential input voltage is driven to zero ($v_+ \simeq v_-$, the “virtual short”), and no current enters the input terminals ($i_+ = i_- = 0$). Master applying these two and most of Chapter 1 becomes mechanical. Full justification: Appendix A.

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Chapter 1

Circuits with the ideal Operational Amplifier

An OPAMP is a differential amplifier, $v_{out} = A_d(v_+ - v_-)$, with a huge but *ill-defined* gain A_d . A useful circuit must therefore *not* depend on the exact value of A_d ; we achieve this with negative feedback. In the ideal limit ($A_d \rightarrow \infty$, input resistance $r_d \rightarrow \infty$, output resistance $r_o \rightarrow 0$) the two golden rules hold and the five standard amplifiers below all follow from them (derivation: Appendix A).

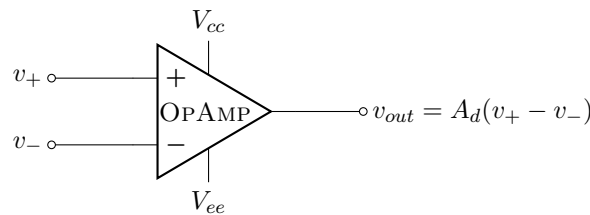


Figure 1.1: The ideal OPAMP symbol. Power rails V_{cc}, V_{ee} are usually omitted in schematics but always present in hardware; the output can never exceed them.

1.1 Designing the five ideal amplifiers

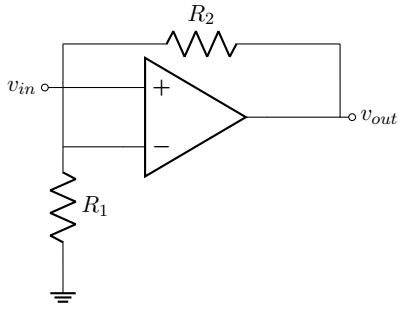
There are five one-OPAMP building blocks (non-inverting and inverting voltage amplifiers, transresistance, transconductance, current amplifier). They are all solved the same way.

Protocol 1.1 Design / analyse an ideal-OpAmp amplifier

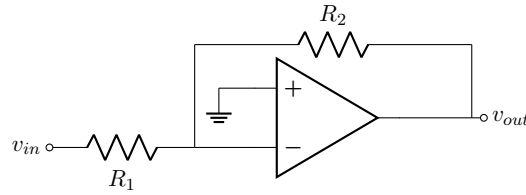
1. **Identify the type** from what is in and what is out: $V \rightarrow V$ (voltage amp), $I \rightarrow V$ (transresistance), $V \rightarrow I$ (transconductance), $I \rightarrow I$ (current amp).
2. **Locate the feedback:** feedback to the $-$ terminal $\Rightarrow$ negative feedback $\Rightarrow$ apply the golden rules.
3. **Apply the rules:** set $v_- = v_+$ (if $+$ is grounded, the $-$ node is a *virtual ground*, $v_- = 0$); set $i_+ = i_- = 0$, so the input current flows entirely through the feedback element.
4. **Write KCL/KVL** at the input node and solve for the transfer function (Table 1.1).
5. **Pick components:** solve the resistor *ratio*, then choose real values from the **E12 series** (Appendix E).
6. **Report the realised value and its error:** recompute the gain with the chosen values; the relative error of a ratio is the sum of the resistors' relative tolerances (Appendix E).

Table 1.1: The five ideal amplifiers (derivations: Appendix A).

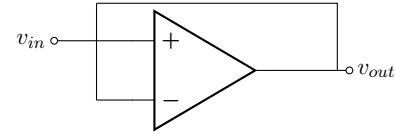
Amplifier	Transfer function	R_{in}	R_{out}
Non-inverting V	$A_V = 1 + \frac{R_2}{R_1}$	∞	0
Inverting V	$A_V = -\frac{R_2}{R_1}$	R_1	0
Transresistance	$v_{out} = -R_2 i_{in}$	0	0
Current	$i_{out} = \left(1 + \frac{R_2}{R_1}\right) i_{in}$	0	∞
Transconductance	$i_{out} = \frac{v_{in}}{R_1}$	∞	∞



(a) Non-inverting



(b) Inverting



(c) Voltage follower

Figure 1.2: The two voltage amplifiers and the unity-gain follower. The follower ($R_1 = \infty, R_2 = 0$) gives $\beta = 1$: the best impedance behaviour at unity gain, used for impedance matching.

Worked Example 1.1 Non-inverting amplifier, $G = 3$

Spec: $A_V = 3$. **Type:** $V \rightarrow V$ non-inverting (Fig. 1.2a). Step 4 gives $A_V = 1 + R_2/R_1 = 3 \Rightarrow R_2 = 2R_1$. One degree of freedom remains. Pick $R_1 = 12 \text{ k}\Omega \Rightarrow R_2 = 24 \text{ k}\Omega$; the nearest E12 value is $22 \text{ k}\Omega$, giving the *realised* gain $G' = 1 + 22/12 = 2.83$ and relative error $\varepsilon_G = |3 - 2.83|/3 = 5.6\%$. If instead exact 12 and $24 \text{ k}\Omega$ resistors at 5% tolerance are used, the gain error is the sum of the two tolerances: $G = 3 \pm 10\%$ (the leading “1” has no uncertainty).

Worked Example 1.2 Transresistance amplifier, current-to-voltage

Spec: input $I_S \in [0, 20 \text{ mA}]$, output $V_{out} \in [0, -5 \text{ V}]$. **Type:** $I \rightarrow V$. The $-$ node is a virtual ground, so i_{in} flows entirely in R_2 and $v_{out} = -R_2 i_{in}$. At the extreme, $-5 = -R_2(0.02) \Rightarrow R_2 = 250 \Omega$. The OPAMP must source/sink the full $i_{out} = 20 \text{ mA}$ — already near typical limits (Chapter 2, Protocol on I_{MAX}).

Exercise 1.1

- (a) Non-inverting $G = 10000$: find β and the loop gain $T = A_d \beta$ for $A_d = 10^5$; comment.
 (b) Non-inverting $G = 0.707$: possible? (c) Inverting $G = -3, -1, -0.707$: give the resistor ratios. (d) Transresistance with $I_S \in [0, 100 \text{ mA}]$ then $[0, 10 \text{ A}]$: what fails?

Solution

- (a) $\beta = 1/G = 10^{-4}$, $T = A_d \beta = 10$ — small, so feedback benefits are weak. (b) **No**: a non-inverting stage cannot give $|A_V| < 1$ ($A_V = 1 + R_2/R_1 \geq 1$); use an *inverting* stage

to attenuate. (c) Inverting $A_V = -R_2/R_1$: ratios 3, 1, 0.707 respectively. (d) Nothing in the *ratio*, but the required output current I_S exceeds what an OPAMP can deliver (100 mA: most cannot; 10 A: none can).

Watch Out

The inverting amplifier's input resistance is only R_1 , *not* ∞ — negative feedback cannot fix it. If the source has resistance R_S , the gain becomes $-R_2/(R_1 + R_S)$; keep $R_1 \gg R_S$.

Chapter 2

The real Operational Amplifier

The ideal model ignores: finite A_d , finite r_d , non-zero r_o , a maximum output current, common-mode gain, and input offsets. We do not re-derive *why* the device is non-ideal; we quantify *how far* from ideal the circuit ends up. Throughout, $\beta = R_1/(R_1 + R_2)$ is the feedback factor and $T = A_d\beta$ the loop gain (Appendix A.1).

2.1 Gain error from finite A_d

With finite A_d the non-inverting gain becomes $A_V = \frac{1}{\beta} \frac{T}{1+T}$: smaller than ideal, by a factor $T/(1+T)$ (derivation: Appendix B).

Protocol 2.1 Gain error due to finite A_d

1. Compute $\beta = 1/A_{V,\text{ideal}}$ (same β for inverting and non-inverting).
2. Compute $T_{\min}, T_{\max} = A_d\beta$ over the datasheet A_d range.
3. Evaluate $A_V = \frac{1}{\beta} \frac{T}{1+T}$ at both ends.
4. Relative uncertainty $\varepsilon_{A_V} = (A_{V,\max} - A_{V,\min})/A_V$.

Worked Example 2.1 $A_V = 10$ with $A_d \in [100\text{k}, 500\text{k}]$

$\beta = 0.1$, so $T_{\min} = 10^4$, $T_{\max} = 5 \times 10^4$. Then $A_{V,\min} = 10 \cdot \frac{T_{\min}}{1+T_{\min}} = 9.9990$ and $A_{V,\max} = 9.99980$, a spread of 0.0008, i.e. $\varepsilon_{A_V} \approx 0.01\%$ — far smaller than the resistor tolerances. Finite A_d is almost never the limiting error.

Exercise 2.1 Does higher gain hurt?

Same device ($A_d \in [100\text{k}, 500\text{k}]$), now at $A_V = 1000$. (a) Find the realised-gain spread. (b) Compare with the $A_V = 10$ case and state the rule.

Solution

(a) $\beta = 10^{-3}$, so $T \in [100, 500]$: $A_{V,\min} = 1000 \cdot \frac{100}{101} = 990.1$ and $A_{V,\max} = 1000 \cdot \frac{500}{501} = 998.0$ — a spread of 7.9, i.e. $\varepsilon_{A_V} \approx 0.8\%$. (b) Dividing the loop gain by 100 multiplied the error by ~ 100 : the gain error scales as $1/T$. Feedback's benefits shrink exactly as fast as you spend loop gain on closed-loop gain — at high gains the finite A_d *can* rival the resistor tolerances.

2.2 Input and output resistance of the amplifier

Negative feedback multiplies the input resistance and divides the output resistance by $(1+T)$: $R_{in} = r_d(1+T)$, $R_{out} = r_o/(1+T)$. Both are found by the *test-source method* (Appendix B.1).

Protocol 2.2 Find R_{in} or R_{out} by the test-source method

1. Kill independent sources (voltage $\rightarrow$ short); keep dependent sources.
2. Apply a test source V_x at the port; find the resulting current I_x .
3. The port resistance is V_x/I_x . Result: $R_{in} = r_d(1 + T)$ (huge), $R_{out} = r_o/(1 + T)$ (tiny).

Worked Example 2.2 Closed-loop impedances of an LM741 stage

LM741 at $A_V = 10$: $r_d = 2 \text{ M}\Omega$, $r_o = 75 \text{ }\Omega$, $A_d = 2 \times 10^5$, so $T = A_d\beta = 2 \times 10^4$. Then $R_{in} = r_d(1 + T) \approx 2 \text{ M} \times 2 \times 10^4 = 40 \text{ G}\Omega$ and $R_{out} = r_o/(1 + T) \approx 75/(2 \times 10^4) = 3.7 \text{ m}\Omega$. Feedback turns a mediocre $2 \text{ M}\Omega/75 \text{ }\Omega$ device into an almost ideal voltage amplifier — as long as T stays large (it falls with frequency, Chapter 3).

Exercise 2.2

The same part wired as a voltage follower. (a) Find R_{in} and R_{out} . (b) Why is the follower the best buffer this device can offer?

Solution

- (a) $\beta = 1 \Rightarrow T = A_d = 2 \times 10^5$: $R_{in} \approx 2 \text{ M} \times 2 \times 10^5 = 400 \text{ G}\Omega$, $R_{out} \approx 75/(2 \times 10^5) = 0.38 \text{ m}\Omega$.
 (b) $\beta = 1$ is the largest possible feedback factor, hence the largest T : the impedance improvement is proportional to the loop gain, and the follower spends *all* of it on impedances (none on amplification).

2.3 Sizing the feedback resistor from I_{MAX}

The output current feeds both the load and the feedback network; keep the feedback share small, $I_{fb} < I_{load}/10$. In the worst case the voltage across R_2 is $2V_{PS}$, giving a *lower bound* $R_2 > \frac{10 \cdot 2V_{PS}}{I_{load}}$.

Protocol 2.3 Size R_2 from I_{MAX}

1. Budget the output current: reserve at most one tenth for the feedback network, $I_{fb} < I_{load}/10$.
2. Worst case the feedback branch sees $2V_{PS}$ (output at one rail, the far end at the other), so $I_{fb} = 2V_{PS}/R_2$.
3. Combine into the *lower* bound $R_2 > \frac{10 \cdot 2V_{PS}}{I_{load}}$.
4. Pair it with the *upper* bound from the offset protocol (§2.6) to get the allowed window for R_2 .

Worked Example 2.3 Lower bound on R_2

$$V_{PS} = 12 \text{ V}, I_{load} = 15 \text{ mA}: R_2 > \frac{10 \cdot 2 \cdot 12}{0.015} = 16 \text{ k}\Omega.$$

Exercise 2.3

$V_{PS} = 15\text{ V}$, $I_{load} = 20\text{ mA}$, non-inverting $A_V = 10$. (a) Bound R_2 and give a resistor pair. (b) What happens to the bound if the load only ever draws 2 mA ?

Solution

(a) $R_2 > \frac{10 \cdot 2 \cdot 15}{0.02} = 15\text{ k}\Omega$. With $A_V = 1 + R_2/R_1 = 10$: e.g. $R_2 = 18\text{ k}\Omega$ (E12), $R_1 = R_2/9 = 2\text{ k}\Omega$. (b) The bound scales as $1/I_{load}$: $R_2 > 150\text{ k}\Omega$. Light loads push R_2 up — straight toward the *upper* bound set by the offset current (§2.6); with both constraints the window can become uncomfortably narrow.

2.4 Common mode and CMRR

Why a common mode exists at all. Any two-input amplifier obeys $v_{out} = A_1 V_1 + A_2 V_2$. A *perfect* differential amplifier would have $A_1 = -A_2$, responding only to the difference of its inputs; a real one does not quite. It is far more revealing to split the inputs into a **differential mode** $v_d = V_1 - V_2$ and a **common mode** $V_C = \frac{1}{2}(V_1 + V_2)$ (their average), which recasts the same output as

$$v_{out} = A_d v_d + A_C V_C, \quad A_d = \frac{A_1 - A_2}{2}, \quad A_C = A_1 + A_2.$$

The differential gain A_d is the one we want; the **common-mode gain** A_C measures the leftover input *asymmetry* — it is zero only if $A_1 = -A_2$ exactly. The figure of merit is the **CMRR** = A_d/A_C (Appendix B.2).

Key Idea

CMRR is not “how much common mode is present” — it is how thoroughly the device *ignores* it. Interference that couples *equally* onto both inputs (e.g. 50 Hz mains hum on a long sensor cable) enters as common mode; a high CMRR rejects it while the wanted difference is amplified normally. Higher CMRR $\Rightarrow$ better differential amplifier.

Protocol 2.4 Use a CMRR (or PSRR) figure

1. From the datasheet CMRR_{dB} recover $\text{CMRR} = 10^{\text{CMRR}_{dB}/20}$.
2. $A_C = A_d/\text{CMRR}$.
3. Insert into the gain expression; with feedback the common-mode term is divided by $\sim 2\text{ CMRR}$ and is normally negligible (Appendix B.2).
4. Treat the **PSRR** (supply rejection) line identically: it is the same ratio taken against a *supply-voltage* change instead of a common-mode input.

Worked Example 2.4 LM741: from a datasheet line to A_C

The LM741 datasheet lists $\text{CMR} = 86\text{ dB}$ (typ.) and a comparable $\text{SVR} = 86\text{ dB}$ (supply-voltage rejection). From the CMR line, $\log_{10}(A_d/A_C) = 86/20 = 4.3$, so $\text{CMRR} = 10^{4.3} = 20\,000$. With the large-signal gain $A_d = 200\text{ V/mV} = 200\,000$,

$$A_C = \frac{A_d}{\text{CMRR}} = \frac{200\,000}{20\,000} = 10.$$

A common-mode gain of 10 *looks* alarming — it equals the wanted gain of an $A_V = 10$ stage. Yet in closed loop the common-mode term carries the factor $1/(2 \text{ CMRR}) = 1/40\,000$, so the transfer function collapses to the ordinary $\frac{1}{\beta} \frac{T}{1+T}$ (Appendix B.2). Negative feedback, once again, hides a large raw imperfection.

Note

Common-mode dynamic range. A high CMRR only helps *inside* the allowed common-mode window: V_C may typically range only over $V_{ee} + 1.5 \text{ V}$ to $V_{cc} - 1.5 \text{ V}$ (down to 0 for single-supply parts, full rail-to-rail for the best ones). Push V_C outside it and the internal stages leave their linear region — the quoted CMRR no longer applies.

Exercise 2.4

(a) A precision OPAMP lists $\text{CMRR}_{dB} = 120 \text{ dB}$ and $A_d = 10^6$: find A_C . (b) For the LM741 stage of the worked example ($\text{CMRR} = 20\,000$), how big is the closed-loop common-mode correction $1/(2 \text{ CMRR})$, compared with 1 % resistors? (c) Sanity-check the exact gain formula of Appendix B.2 in the limit $\text{CMRR} \rightarrow \infty$.

Solution

(a) $\text{CMRR} = 10^{120/20} = 10^6$, so $A_C = A_d/\text{CMRR} = 1$. (b) $1/(2 \cdot 20\,000) = 2.5 \times 10^{-5} = 0.0025 \%$ — about 400 times below a single 1 % resistor tolerance; CMRR is never the accuracy bottleneck of this stage. (c) Setting $1/(2 \text{ CMRR}) \rightarrow 0$ collapses the exact expression to $\frac{1}{\beta} \frac{T}{1+T}$ — the finite- A_d result with no common-mode error, as it must.

2.5 Output voltage swing versus load

The peak output is the *smaller* of the supply-imposed limit and the current-imposed limit $V_p = I_{MAX} R_L$. For small R_L the current limit wins; above $R_L = V_{MAX}/I_{MAX}$ the supply limit dominates.

Protocol 2.5 Output swing into a load

1. Current limit: $V_p^{(I)} = I_{MAX} R_L$.
2. Supply limit: $V_p^{(V)} = V_{MAX}$ (the rail minus the output stage's drop).
3. The achievable peak is the *smaller* of the two; the crossover load is $R_L^* = V_{MAX}/I_{MAX}$ — below it you are current-limited.

Worked Example 2.5 Swing for several loads, $I_{MAX} = 20 \text{ mA}$

$R_L = 0.1 \text{ k}\Omega$: $V_p = I_{MAX} R_L = 2 \text{ V}$ ($V_{pp} = 4 \text{ V}$). $R_L = 0.3 \text{ k}\Omega$: $V_{pp} = 12 \text{ V}$; $R_L = 0.5 \text{ k}\Omega$: $V_{pp} = 20 \text{ V}$. The crossover to supply-limited swing (for $V_{MAX} = 15 \text{ V}$) is $R_L = 15/0.02 = 0.75 \text{ k}\Omega$.

Exercise 2.5

$I_{MAX} = 25 \text{ mA}$, $V_{MAX} = 10 \text{ V}$. Find the peak swing into (a) $R_L = 200 \Omega$ and (b) $R_L = 1 \text{ k}\Omega$, and (c) the crossover load.

Solution

(a) $V_p = \min(0.025 \times 200, 10) = 5 \text{ V}$ — current-limited. (b) $V_p = \min(25, 10) = 10 \text{ V}$ — supply-limited. (c) $R_L^* = 10/0.025 = 400 \Omega$.

2.6 Output offset and how to minimise it

Three imperfections create a non-zero output even with zero input: the input offset voltage V_{OFF} , the bias current I_{BIAS} , and the offset current I_{OFF} . By superposition (Appendix B.3), and after adding a resistor R_3 on the $+$ terminal,

$$v_{out,OFF} = V_{OFF} \left(1 + \frac{R_2}{R_1} \right) + I_{BIAS} \left[R_2 - R_3 \left(1 + \frac{R_2}{R_1} \right) \right] + I_{OFF} R_2.$$

Protocol 2.6 Minimise the output offset

1. **Cancel the bias term:** set $R_3 = R_1 \parallel R_2$ (more generally, equal resistance seen from each input). The I_{BIAS} term vanishes.
2. What remains: $v_{out,OFF} = V_{OFF}(1 + R_2/R_1) + I_{OFF}R_2$.
3. **Bound R_2 :** make the offset-current term small versus the offset-voltage term: $R_2 \ll \frac{V_{OFF}}{I_{OFF}} A_V$.
4. Combine with the *lower* bound from I_{MAX} to get a valid range for R_2 . If V_{OFF} still dominates, pick a better OPAMP.

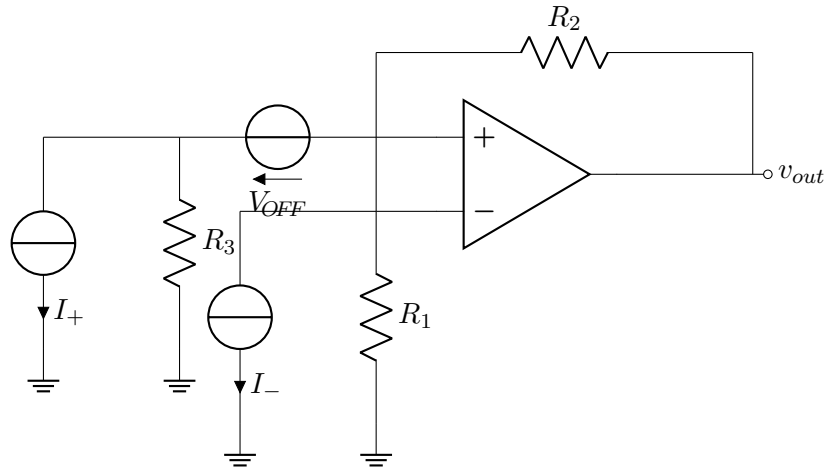


Figure 2.1: Offset model: input bias/offset currents (I_+ , I_-) and input offset voltage V_{OFF} added to an otherwise ideal OPAMP; R_3 on the $+$ terminal cancels the bias-current contribution when $R_3 = R_1 \parallel R_2$.

Worked Example 2.6 Upper bound on R_2

$A_V = 10$, $V_{OFF} = 5 \text{ mV}$, $I_{OFF} = 60 \text{ nA}$: $R_2 \ll \frac{5 \times 10^{-3}}{60 \times 10^{-9}} \cdot 10 = 83 \text{ k}\Omega$. With the earlier lower bound (e.g. $16 \text{ k}\Omega$), a sensible window for R_2 is roughly $16 \text{ k}\Omega$ to $83 \text{ k}\Omega$.

Exercise 2.6 A full R_2 window

$A_V = 20$, $V_{OFF} = 2 \text{ mV}$, $I_{OFF} = 20 \text{ nA}$, $V_{PS} = 12 \text{ V}$, $I_{load} = 10 \text{ mA}$. (a) Give the window for R_2 and pick a value. (b) With $R_2 = 100 \text{ k}\Omega$ and $R_3 = R_1 \parallel R_2$ in place, estimate the residual output offset.

Solution

(a) Lower bound (I_{MAX}): $R_2 > 10 \cdot 2 \cdot 12 / 0.01 = 24 \text{ k}\Omega$. Upper bound: $R_2 \ll \frac{V_{OFF}}{I_{OFF}} A_V = \frac{2 \times 10^{-3}}{2 \times 10^{-8}} \cdot 20 = 2 \text{ M}\Omega$. $R_2 = 100 \text{ k}\Omega$ keeps a decade of margin on both sides. (b) $v_{out,OFF} = V_{OFF} A_V + I_{OFF} R_2 = 2 \text{ mV} \times 20 + 2 \times 10^{-8} \times 10^5 = 40 + 2 = 42 \text{ mV}$ — the offset *voltage* dominates, exactly what the upper bound guaranteed.

Chapter 3

Frequency response, stability and bandwidth

Everything so far assumed DC. Real signals (tones, square waves, voice) have a spectrum, and the OPAMP's open-loop gain A_d is *not* constant with frequency. Internally the device is three cascaded stages — a differential input stage, a high-gain intermediate stage and an output stage — and each stage adds one **pole** from its parasitic capacitance. So $A_d(f)$ keeps its large DC value A_{d0} at low frequency but rolls off through (typically) *three* poles, the first two more than a decade apart. The phase those poles add can turn the negative feedback *positive* and make the amplifier oscillate, so we (i) build the loop gain, (ii) check **stability**, (iii) **compensate** if needed, then (iv) read off the **bandwidth**. Bode construction and the closed-loop derivation: Appendix C.

3.1 The loop gain and what each pole does

One pole bends the asymptotic magnitude down by 20 dB per decade and rotates the phase by -90° — the rotation starts a decade before the pole, is -45° at the pole, and completes a decade after it. If two poles are closer than two decades their phase shifts overlap and add.

The feedback network β is normally just resistors, so it adds *no* poles and *no* phase: on a Bode plot $|\beta|$ is a flat negative constant ($\beta = 1 \Rightarrow 0$ dB for the follower; $A_V = 5 \Rightarrow \beta = 1/5$, i.e. -14 dB). The quantity that decides stability is the **loop gain** $T(f) = A_d(f)\beta$: its magnitude is $|A_d|$ shifted *down* by $|\beta|_{\text{dB}}$, and its phase is identical to that of A_d .

Key Idea

Stability is read on $T = A_d\beta$, not on A_d alone. The poles live in A_d (the device); β (your resistors) only slides the magnitude up or down without touching the phase. Lowering the closed-loop gain raises β , lifts $|T|$, and pushes the 0 dB crossing to a higher — more dangerous — frequency, so the **voltage follower** ($\beta = 1$) is the worst case.

3.2 Phase margin: the stability test

The signal returns to the inverting input, so the total loop phase shift is $\phi_{FB} = 180^\circ + \angle T$. Feedback stays negative while ϕ_{FB} is far from 0° ; where the poles drag $\angle T \rightarrow -180^\circ$ the feedback turns positive and oscillation is possible. The safety figure is the **phase margin**

$$\phi_M = 180^\circ + \angle T|_{|T|=0 \text{ dB}},$$

the phase left when the loop gain has fallen to 0 dB. A design wants $\phi_M \geq 45^\circ$; since a pole eats 45° at its own frequency, $\phi_M = 45^\circ$ is exactly the condition “ $|T| = 0$ dB at the second pole f_{p2} ”. (Equivalently the *gain margin* — $|T|$ measured where $\angle T = -180^\circ$ — must be below 0 dB.)

Protocol 3.1 Analyse stability

1. Get $A_d(f)$ from the datasheet: DC gain A_{d0} and poles $f_{p1}, f_{p2}, \dots$
2. Find β from the feedback network ($\beta = 1/A_V$). An inverting stage uses the *same* network, so treat it as a non-inverting stage of gain $1/\beta$. Draw $|T| = |A_d| - |\beta|_{\text{dB}}$.
3. Read ϕ_M where $|T| = 0$ dB. If $\phi_M \geq 45^\circ$ (the crossing is at or below f_{p2}) it is stable — stop.
4. Otherwise **compensate** (next section) so the crossing moves down to f_{p2} , then recompute $A_V(f)$ and the bandwidth.

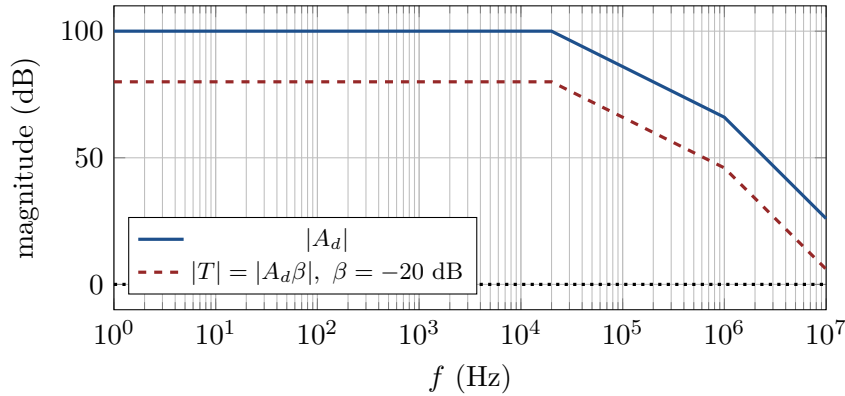


Figure 3.1: Asymptotic Bode magnitudes for $A_{d0} = 100$ dB, poles at 20 kHz and 1 MHz, with $\beta = -20$ dB ($A_V = 10$). Stability/compensation act to bring $|T|$ to 0 dB no later than the second pole.

Worked Example 3.1 Reading the phase margin off the Bode plot

The running device ($A_{d0} = 100$ dB, $f_{p1} = 20$ kHz, $f_{p2} = 1$ MHz) at $A_V = 100$: $|\beta| = -40$ dB, so $|T_0| = 60$ dB. From f_{p1} , $|T|$ falls at 20 dB per decade and reaches $60 - 20 \log_{10}(10^6/2 \times 10^4) = 60 - 34 = 26$ dB at f_{p2} — still above 0 dB. The crossing therefore happens *beyond* f_{p2} (about $26/40 = 0.65$ decades past it, near 4.5 MHz), where the second pole has already dragged the phase below -135° : $\phi_M < 45^\circ$, **unstable** — compensation required.

Exercise 3.1

Same device, no compensation allowed. (a) At which closed-loop gain does the stage become *just* stable ($\phi_M = 45^\circ$)? (b) Check $A_V = 5000$ explicitly.

Solution

(a) Just stable $\Leftrightarrow |T| = 0$ dB exactly at f_{p2} , i.e. $|\beta|_{\text{dB}} = -|A_d(f_{p2})| = -66$ dB: $A_V = 10^{66/20} \approx 2000$. Any larger gain is stable too (smaller β lowers $|T|$ further). (b) $A_V = 5000 \Rightarrow |\beta| = -74$ dB, $|T_0| = 26$ dB: $|T|$ reaches 0 dB after $26/20 = 1.3$ decades, at $2 \times 10^4 \times 10^{1.3} \approx 400$ kHz $< f_{p2}$ — crossing before the second pole, $\phi_M > 45^\circ$, stable. High-gain stages are the easy case; the follower is the dangerous one.

3.3 Four ways to compensate

If the stage is unstable we reshape the loop so $|T|$ reaches 0 dB no later than f_{p2} . There are three textbook routes — the second has two variants, hence “2.1” and “2.2” — all acting on the device gain A_d , because β is fixed by the wanted closed-loop gain and may not be touched. They differ sharply in the two things you care about: the surviving **DC loop gain** (how much feedback benefit is left) and the resulting **bandwidth**.

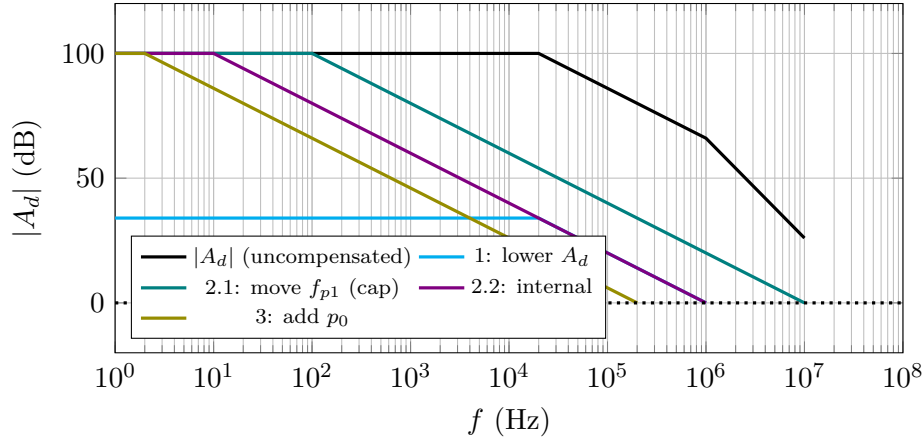


Figure 3.2: The four compensated open-loop gains for the running example ($A_{d0} = 100$ dB, $f_{p1} = 20$ kHz, $f_{p2} = 1$ MHz). Here the curves are the compensated $|A_d|$; the loop gain $|T|$ is each curve shifted down by $|\beta| = 20$ dB, so the bandwidth is where $|A_d| = 20$ dB. Method 2.1 keeps the highest curve longest (best bandwidth); method 3 rolls off first (worst); method 1 also throws the DC gain away.

Protocol 3.2 Compensate the OpAmp

1. Confirm the uncompensated stage is unstable ($\phi_M < 45^\circ$).
2. If you control the board *and* the part exposes compensation pins, use **Method 2.1**: move f_{p1} to f_{p2}/T_0 — best bandwidth.
3. Otherwise choose an **internally compensated** part (**Method 2.2**): f_{p1} is fixed at f_{p2}/A_{d0} , stable for any $A_V \geq 1$, and the bandwidth is f_{BW}/A_{V0} .
4. Avoid **Method 1** (it guts the loop gain) and **Method 3** (it guts the bandwidth) unless forced.
5. Recompute $A_V(f)$ and verify the bandwidth via the GBWP (next section).

Note

Running example (used in all four methods below). A non-inverting stage of gain $A_V = 10$ ($\beta = 1/10$, $|\beta| = -20$ dB) on an OPAMP with $A_{d0} = 100$ dB (10^5), $f_{p1} = 20$ kHz, $f_{p2} = 1$ MHz. The poles are $n_{\text{dec}} = \log_{10}(f_{p2}/f_{p1}) = \log_{10} 50 = 1.7$ decades apart, so $|A_d(f_{p2})| = 100 - 20 \cdot 1.7 = 66$ dB; the DC loop gain is $T_0 = A_{d0} \beta = 10^4$, i.e. $|T_0| = 80$ dB and $|T(f_{p2})| = 80 - 34 = 46$ dB. Uncompensated, $|T|$ crosses 0 dB well past f_{p2} — unstable.

3.3.1 Method 1 — lower the differential gain A_d

Slide the whole $|A_d|$ curve down (without moving any pole) until $|T(f_{p2})| = 0$ dB. The required attenuation is $M = |T(f_{p2})|$.

Worked Example 3.2 Method 1 on the running example

$M = |T(f_{p2})| = 46$ dB, so the DC loop gain collapses from 80 dB to $80 - 46 = 34$ dB — already a poor return on the feedback. Worse, if the manufacturer does this inside the IC it must assume the worst case $\beta = 1$, so $M = |A_d(f_{p2})| = 66$ dB and A_{d0} drops to 34 dB; used at $A_V = 10$ the loop gain is then only $34 - 20 = 14$ dB. Stable, but the feedback benefits (gain accuracy, low distortion, better R_{in}/R_{out}) are almost gone.

Exercise 3.2

The same OPAMP ($A_{d0} = 100$ dB, $f_{p1} = 20$ kHz, $f_{p2} = 1$ MHz) used as a *voltage follower*. Worst-case Method 1: find M , the compensated A_{d0} , and the frequency where $|T|$ now crosses 0 dB.

Solution

Follower $\Rightarrow \beta = 1$, so $T = A_d$. $M = |A_d(f_{p2})| = 66$ dB, giving $A_{d0} = 100 - 66 = 34$ dB (≈ 50). The pole is still at $f_{p1} = 20$ kHz, so $|T| = 0$ at $f_{p1} \cdot 10^{34/20} = 20 \text{ k} \times 50 = 1 \text{ MHz} = f_{p2}$, i.e. $\phi_M = 45^\circ$ — just stable. But the DC loop gain is now only 34 dB: the method works yet throws feedback away, so in practice it is the last choice.

3.3.2 Method 2.1 — move the first pole with an external capacitor

Add a capacitor on the dedicated compensation pins to drag f_{p1} down (leaving f_{p2}, f_{p3} in place) until $|T| = 0$ dB lands exactly on f_{p2} . Because you know *your* β , you move the pole only as far as this gain needs — the highest possible bandwidth, $f_T = f_{p2}$. The new pole sits $|T_0|_{\text{dB}}/20$ decades below f_{p2} :

$$f_{p1}^{\text{new}} = f_{p2}/10^{|T_0|_{\text{dB}}/20} = f_{p2}/T_0.$$

Worked Example 3.3 Method 2.1 on the running example

$f_{p1}^{\text{new}} = f_{p2}/10^{80/20} = 10^6/10^4 = 100$ Hz. The DC loop gain stays at the full 80 dB and the closed-loop bandwidth is the best attainable, $f_T = f_{p2} = 1$ MHz. The price: an external capacitor and pins that are not always available.

Exercise 3.3

The same OPAMP, now at $A_V = 100$. Where must Method 2.1 place f_{p1}^{new} , and what DC loop gain survives?

Solution

$\beta = 1/100$, $|T_0| = 100 - 40 = 60$ dB (10^3). Move the pole to $f_{p1}^{\text{new}} = f_{p2}/10^{60/20} = 10^6/10^3 = 1$ kHz. The DC loop gain is the full 60 dB — Method 2.1 always keeps it; that is its appeal.

3.3.3 Method 2.2 — use an internally compensated OpAmp

When you cannot add a capacitor, pick a part the manufacturer has already compensated for the *worst case* $\beta = 1$ (the follower): the internal cap moves f_{p1} until $|A_d(f_{p2})| = 0$ dB, i.e.

$$f_{p1}^{new} = f_{p2}/10^{|A_{d0}|_{dB}/20} = f_{p2}/A_{d0}.$$

It is then stable for *any* $A_V \geq 1$. This is the everyday choice (LM741, TL08x, ...): trivial to use and the full DC loop gain is kept; only the bandwidth shrinks as gain rises.

Worked Example 3.4 Method 2.2 on the running example, plus a datasheet

$f_{p1}^{new} = f_{p2}/A_{d0} = 10^6/10^5 = 10$ Hz, lower than Method 2.1's 100 Hz (because it must also cover the follower) — hence less bandwidth, but unconditionally stable. From a real datasheet: $A_{d0} = 400$ k (112 dB), $f_{p2} = 1$ MHz $\Rightarrow f_{p1}^{new} = 10^6/(400 \times 10^3) = 2.5$ Hz, and the measured phase margin is $\phi_M = 90^\circ$ — well past the 45° minimum (and even larger at $A_V > 1$).

Exercise 3.4

The same internally compensated part is used once as a follower and once at $A_V = 10$. Does f_{p1}^{new} change between the two uses? What *does* change?

Solution

No — $f_{p1}^{new} = f_{p2}/A_{d0}$ is fixed in the factory for the worst case $\beta = 1$, so it is the same part however you wire it. What changes is the **bandwidth**: $f_T = f_{BW}/A_{V0}$, so the follower gets the full $f_T = f_{BW} = f_{p2}$ while the $A_V = 10$ stage gets only $f_{p2}/10$. Higher gain always costs bandwidth here — that is the GBWP (next section).

3.3.4 Method 3 — add a new dominant pole p_0

Introduce a brand-new low-frequency pole p_0 below f_{p1} so it becomes dominant and the old f_{p1} slips into the role of “second pole”. Place it so $|T|$ is already at 0 dB by the *original* first pole:

$$f_{p0} = f_{p1}/10^{|T_0|_{dB}/20} = f_{p1}/T_0.$$

Because the rolloff now starts so early, this gives the *worst* bandwidth of all.

Worked Example 3.5 Method 3 on the running example

$f_{p0} = f_{p1}/T_0 = 20 \text{ k}/10^4 = 2$ Hz. The DC loop gain is kept (80 dB, good feedback), but the bandwidth is crushed to $f_T = f_{p1} = 20$ kHz — two decades below Method 2.1. Rarely used for exactly this reason.

Exercise 3.5

The same OPAMP at $A_V = 100$. Where does Method 3 place p_0 ? Where would Method 2.1 instead put the moved first pole, and why does Method 3 leave so much less usable bandwidth?

Solution

$|T_0| = 60 \text{ dB}$ (10^3): Method 3 sets $f_{p0} = f_{p1}/10^3 = 20 \text{ k}/1000 = 20 \text{ Hz}$, while Method 2.1 would put the moved pole at $f_{p2}/10^3 = 1 \text{ kHz}$. Both keep the full 60 dB of DC loop gain, but Method 3's dominant pole sits a factor $f_{p1}/f_{p2} = 1/50$ lower, so its closed-loop bandwidth is about $50\times$ smaller. Method 3 protects the feedback but pays the heaviest bandwidth penalty.

The four methods at the running gain $A_V = 10$ side by side:

Method	New dominant pole	DC loop gain T_0	Bandwidth f_T	When to use
1 lower A_d	f_{p1} unmoved (20 kHz)	14 dB (guttled)	100 kHz	avoid
2.1 external cap	100 Hz	80 dB (full)	1 MHz	best, if pins exist
2.2 internal	10 Hz	80 dB (full)	100 kHz	default / easiest
3 add p_0	2 Hz	80 dB (full)	20 kHz	rare (worst BW)

Note how Methods 1 and 2.2 reach the *same* bandwidth (100 kHz) yet Method 2.2 keeps 80 dB of loop gain against Method 1's 14 dB: that is why nobody uses Method 1.

Note

Inverting stages. An inverting amplifier has the same feedback network as the non-inverting one, hence the same $\beta = R_1/(R_1 + R_2)$ and the same loop gain $T = A_d\beta$ — stability does not depend on where V_{in} connects. Analyse it as a non-inverting stage of gain $1/\beta$. Example: $A_V = -2$ needs $R_2/R_1 = 2$, so $\beta = \frac{R_1}{R_1+R_2} = \frac{1}{3}$ and you study the stability of a non-inverting stage of gain $1/\beta = 3$.

3.4 Bandwidth from the Gain–Bandwidth Product

Take an internally compensated part (Method 2.2): a single dominant pole at f_0 makes the closed loop look like one pole,

$$A_V(f) \simeq \frac{1}{\beta} \frac{1}{1 + jf/(A_d0\beta f_0)}, \quad f_T = A_d0\beta f_0 = \frac{f_{BW}}{A_{V0}},$$

with $f_{BW} = A_d0f_0$ a *constant* of the device (Appendix C.1). So gain and bandwidth trade one-for-one: doubling the gain halves the bandwidth. The follower ($A_{V0} = 1$) gets the whole f_{BW} . (The GBWP shortcut applies only to internally compensated parts; with Method 2.1 the cap is retuned per gain and $f_T = f_{p2}$ always.)

Protocol 3.3 Bandwidth via GBWP

1. $f_T = \text{GBWP}/A_{V0}$ (for an inverting stage use the equivalent non-inverting gain $A_{V0} = |A_{inv}| + 1$).
2. If $f_T \geq f_C$ (the required bandwidth), done.
3. Else either pick a faster OPAMP ($\text{GBWP} \geq A_{V0}f_C$), *or* split the gain across cascaded stages so each stage keeps a higher f_T .

Worked Example 3.6 GBWP, pass and fail

GBWP = 1 MHz, want $f_C = 40$ kHz. (1) $A_{V0} = 12 \Rightarrow f_T = 10^6/12 = 83.3$ kHz $> f_C$ — meets spec. (2) $A_{V0} = 100 \Rightarrow f_T = 10$ kHz $< f_C$ — fails. Fix it by choosing a part with GBWP $\geq 100 \times 40$ k = 4 MHz, *or* by cascading two $\times 10$ stages: each has $f'_T = 10^6/10 = 100$ kHz, the product gain is 100, and the combined bandwidth (slightly under 100 kHz) clears the spec.

Exercise 3.6

A non-inverting stage needs $A_{V0} = 25$ and a bandwidth $f_C = 50$ kHz from a part with GBWP = 1 MHz. Does it meet spec? Give two fixes.

Solution

$f_T = 10^6/25 = 40$ kHz < 50 kHz — **fails**. (i) Use a faster OPAMP: GBWP $\geq 25 \times 50$ k = 1.25 MHz. (ii) Cascade two $\times 5$ stages: each $f'_T = 10^6/5 = 200$ kHz, product gain 25, combined bandwidth just under 200 kHz — spec met with margin.

3.5 Slew rate

Bandwidth is a *small-signal* limit. The **slew rate** is the *large-signal* ceiling on how fast the output can move,

$$\text{SR} = \max \frac{dv_{out}}{dt} = \frac{I_s}{C_c},$$

set by the largest current I_s the OPAMP can push into its compensation capacitor C_c ($\Delta V_C = \frac{1}{C_c} \int i dt = \frac{I_s}{C_c} t$ for a constant current). Feed an ideal step and the output ramps at this fixed slope instead of following.

Note

Bandwidth vs slew rate on a step. Both come from the same internal capacitances but answer different questions. The small-signal rise time $t_{10-90} = 2.2\tau = 0.35/\text{BW}$ (with $\tau = 1/2\pi \text{ BW}$) is set by the bandwidth and is *independent* of the step height. The slewing rise time $t_{10-90} = 0.8 V_H/\text{SR}$ *grows* with the step height V_H . Small steps are bandwidth-limited; large steps are slew-limited.

For a sinusoid $v_{out} = V_p \sin \omega t$ the steepest slope is $V_p \omega$, so the output stays undistorted only while

$$V_p \omega \leq \text{SR} \implies V_p \leq \frac{\text{SR}}{2\pi f} \quad (\text{full-power bandwidth}).$$

The usable amplitude therefore falls off as $1/f$; at low frequency it is capped instead by the output dynamic range.

Protocol 3.4 Bandwidth vs slew-rate limit

1. Max gain from GBWP: $G_{\max} = \text{GBWP}/f_0$.
2. Max undistorted amplitude from SR: $V_p < \text{SR}/(2\pi f_0)$.
3. Whichever bound bites first — gain (bandwidth) or amplitude (slew) — limits the design; if both are respected there is no limitation.

Worked Example 3.7 Full-power bandwidth of an LM741

$SR = 0.5 \text{ V}/\mu\text{s} = 500 \text{ kV/s}$. The largest *undistorted* frequency depends on the amplitude: $f = SR/(2\pi V_p)$. For the full $\pm 12 \text{ V}$ swing $f = 5 \times 10^5/(2\pi \cdot 12) = 6.63 \text{ kHz}$; for $\pm 10 \text{ V}$, 7.96 kHz ; $\pm 5 \text{ V}$, 15.9 kHz ; $\pm 1 \text{ V}$, 79.6 kHz . Smaller signals can run faster.

Worked Example 3.8 TL082: which limit bites?

$SR = 13 \text{ V}/\mu\text{s}$, $GBWP = 4 \text{ MHz}$, $f_0 = 400 \text{ kHz}$. $G_{\max} = 4 \times 10^6/4 \times 10^5 = 10$; $V_p < 13 \times 10^6/(2\pi \cdot 4 \times 10^5) = 5.17 \text{ V}$. So: $V_p < 5.17 \text{ V}$ but gain $> 10 \Rightarrow$ bandwidth limits; gain < 10 but $V_p > 5.17 \text{ V} \Rightarrow$ slew rate limits; both respected $\Rightarrow$ no limitation.

Exercise 3.7

An LM741 ($SR = 0.5 \text{ V}/\mu\text{s}$) must output a 10 kHz sine. (a) What is the largest amplitude V_p free of slewing? (b) If a 10 V peak is required, above what frequency does it slew?

Solution

(a) $V_p \leq SR/(2\pi f) = 5 \times 10^5/(2\pi \cdot 10^4) = 7.96 \text{ V}$. (b) $f \leq SR/(2\pi V_p) = 5 \times 10^5/(2\pi \cdot 10) = 7.96 \text{ kHz}$, so a 10 V peak already slews at 10 kHz ; drop the amplitude or use a faster part (e.g. a TL082).

Chapter 4

Applications

4.1 The audio amplifier (placing poles and zeros)

A practical amplifier has a finite pass band $[f_L, f_H]$ and a chosen input impedance. The non-inverting stage gets three capacitors: C_3 blocks input DC, C_1 sets the DC gain to 1 (kills offset amplification), C_2 rolls off the high end. Analyse it at three frequencies (Appendix C).

Protocol 4.1 Design the audio amplifier

1. **DC** (caps open): input disconnected, gain to offset is 1 — good.
2. **Pass band** (C_1, C_3 short, C_2 open): $R_{in} = R_3$; set $R_3 = R_{in}^{spec}$, then $R_2 = R_3$ (offset), then $R_1 = R_2/(A_{V0} - 1)$.
3. **High end**: pole $f_H = 1/(2\pi R_2 C_2) \Rightarrow C_2$.
4. **Low end**: poles $f_{p1} = 1/(2\pi R_1 C_1) = f_L$ and $f_{p3} = 1/(2\pi R_3 C_3)$ (place a decade below f_L) $\Rightarrow C_1, C_3$.

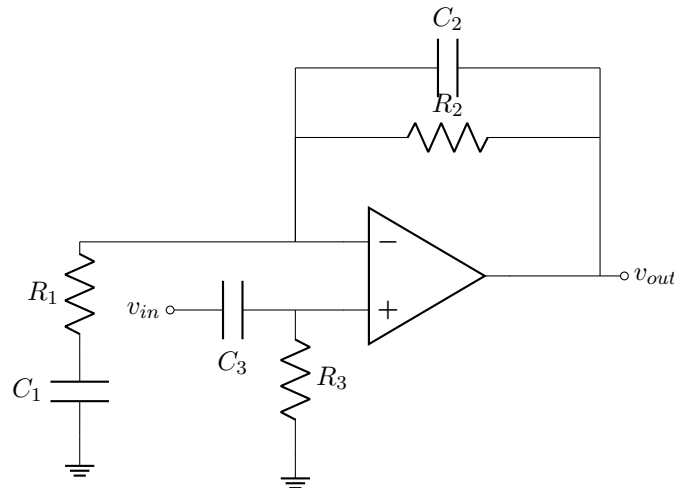


Figure 4.1: Audio amplifier: C_3 AC-couples the input, C_1 forces unity DC gain, C_2 sets the upper cutoff.

Worked Example 4.1 Full audio design

$f_L = 20$ Hz, $f_H = 20$ kHz, $A_{V0} = 10$, $R_{in} = 100$ k Ω . Then $R_3 = R_2 = 100$ k Ω , $R_1 = R_2/(A_{V0} - 1) = 100/9 \approx 12$ k Ω . Caps: $C_2 = 1/(2\pi R_2 f_H) = 80$ pF, $C_1 = 1/(2\pi R_1 f_L) = 663$ nF, $C_3 = 1/(2\pi R_3 f_{p3})$ with $f_{p3} = 2$ Hz $\Rightarrow 800$ nF.

Exercise 4.1

Design the same topology for $f_L = 50$ Hz, $f_H = 15$ kHz, $A_{V0} = 20$, $R_{in} = 47$ k Ω (place f_{p3} a decade below f_L).

Solution

$R_3 = R_2 = 47$ k Ω ; $R_1 = R_2/(A_{V0} - 1) = 47/19 \approx 2.5$ k Ω (nearest E12 2.7 k Ω realises $A_{V0} = 18.4$; keep 2.5 k Ω if the gain matters). Caps: $C_2 = 1/(2\pi R_2 f_H) = 226$ pF $\rightarrow$ 220 pF; $C_1 = 1/(2\pi R_1 f_L) = 1.27$ μ F $\rightarrow$ 1.2 μ F; $f_{p3} = 5$ Hz: $C_3 = 1/(2\pi R_3 f_{p3}) = 677$ nF $\rightarrow$ 680 nF.

4.2 Generalised adder/subtractor

To realise $v_{out} = \sum A'_j v'_j - \sum A''_i v''_i$, feed the “+” inputs to the non-inverting terminal and the “−” inputs to the inverting terminal, each through $R = R_f/A$.

Protocol 4.2 Generalised adder/subtractor

1. Fix R_f . For every coefficient set its resistor $R = R_f/|A|$ (− signs $\rightarrow$ inverting side, + signs $\rightarrow$ non-inverting side).
2. Balance: require $(1 + \sum A''_i)/\sum A'_j = 1$. If > 1 add a resistor to ground on the + side; if < 1 add one on the − side. This also equalises the input resistances (offset minimised).

Worked Example 4.2 $v_{out} = 2v'_1 + 5v'_2$

$\sum A'_j = 7$, $\sum A''_i = 0$, so $(1 + 0)/7 < 1$: add a resistor on the − side with $A''_1 = 6$. Resistors: $R'_1 = R_f/2$, $R'_2 = R_f/5$, $R''_1 = R_f/6$. Both input terminals then see $R_f/7$ — offsets minimised.

Exercise 4.2

Realise (a) $v_{out} = -2v_1 - 3v_2 - 4v_3$ and (b) the difference $v_{out} = 6(v'_1 - v''_1)$. Give the resistors and the balancing element.

Solution

(a) All inverting: $R''_i = R_f/2, R_f/3, R_f/4$; here $(1 + 9)/0$ diverges, so add a +-side resistor with $A'_1 = 10$, $R'_1 = R_f/10$ (the + node is not simply grounded). (b) $A'_1 = A''_1 = 6$: $R'_1 = R''_1 = R_f/6$; $(1 + 6)/6 > 1 \Rightarrow$ add $A'_2 = 1$, $R'_2 = R_f$ to ground on the + side. It is a differential amplifier.

4.3 The integrator

Replace the inverting amp’s feedback resistor with a capacitor: $v_{out}(t) = -\frac{1}{RC} \int v_{in} dt$ (pole at the origin). A constant input gives a ramp; a square wave gives a triangle of peak-to-peak $V_{in}(T/2)/RC$; a sine comes out scaled by $1/(\omega RC)$ and shifted by $+90^\circ$ (on top of the inversion).

Why the textbook integrator fails as drawn. At DC the capacitor is an open circuit, so there is *no* feedback at all: the input offset voltage and bias current are integrated unopposed and the output drifts (“winds up”) to a rail within seconds. The fix is a resistor R_2 in parallel with C : it restores

DC feedback with gain $-R_2/R$, turning the pole at the origin into a pole at $f_p = 1/(2\pi R_2 C)$. The circuit then integrates only for $f \gg f_p$ — which is all a real application needs.

Protocol 4.3 Design a practical integrator

1. Set RC from the job: triangle from a square wave, $V_{pp,tri} = V_{in} \frac{T/2}{RC}$; or gain at the working frequency, $|A(f)| = 1/(2\pi f RC)$.
2. Split RC : choose R well above the source resistance, then $C = RC/R$.
3. Add $R_2 = A_{DC} R$ across C , with A_{DC} the largest tolerable DC (offset) gain; check that the pole $f_p = 1/(2\pi R_2 C)$ sits at least a decade below the lowest working frequency.
4. Check the residual DC output, $\approx V_{OFF}(1 + R_2/R) + I_{BIAS}R_2$ (and add $R_3 = R \parallel R_2$ on the + input as usual, §2.6).

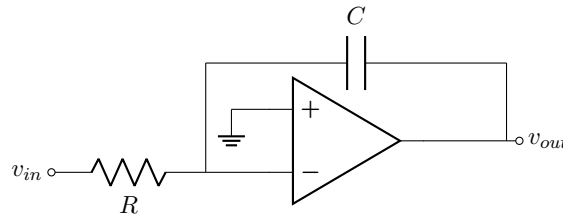


Figure 4.2: Inverting integrator.

Worked Example 4.3 Square wave $\rightarrow$ triangle

Input: ± 1 V square wave at 1 kHz; wanted: a 2 V peak-to-peak triangle. During each half period ($T/2 = 0.5$ ms) the constant ± 1 V input ramps the output by $V_{in}(T/2)/RC$, so $2 = 1 \times 0.5 \text{ ms}/RC \Rightarrow RC = 0.25 \text{ ms}$. Pick $C = 10 \text{ nF} \Rightarrow R = 25 \text{ k}\Omega$ (E12 27 k Ω gives $V_{pp} = 1.85 \text{ V}$, -7%). DC stability: accept $A_{DC} = 100$, so $R_2 = 100R = 2.5 \text{ M}\Omega$ and $f_p = 1/(2\pi R_2 C) = 6.4 \text{ Hz}$ — more than two decades below 1 kHz, so the triangle is unaffected.

Exercise 4.3

- (a) Feed the integrator above ($RC = 0.25 \text{ ms}$) a 100 mV sine at 500 Hz: output amplitude and phase? (b) Remove R_2 and take $I_{BIAS} = 100 \text{ nA}$: starting from 0, how long until the output hits a 12 V rail?

Solution

(a) $|A| = 1/(2\pi f RC) = 1/(2\pi \times 500 \times 2.5 \times 10^{-4}) = 1.27$: output 127 mV, at $-90^\circ + 180^\circ = +90^\circ$ (integration plus inversion). (b) With no DC feedback I_{BIAS} flows straight into C : $dv/dt = I/C = 10^{-7}/10^{-8} = 10 \text{ V/s}$, so the rail arrives in $\approx 1.2 \text{ s}$ — the wind-up that R_2 exists to stop.

Chapter 5

The Operational Amplifier as a comparator

Without negative feedback the OPAMP saturates: it becomes a **comparator**, output V_{OH} or V_{OL} . This chapter climbs the whole ladder: the bare comparator and why noise makes it chatter (§5.1), the two Schmitt-trigger topologies that fix it (§5.2–5.3), and the two oscillators built from them — the astable square-wave generator and the triangle/square generator.

5.1 The bare threshold comparator

Open the loop and the OPAMP becomes a one-bit analog-to-digital converter: with $A_d \sim 10^5$, a differential input of a few tens of μV already rails the output, so for any practical input

$$v_{out} = \begin{cases} V_{OH} & v_+ > v_- , \\ V_{OL} & v_+ < v_- . \end{cases}$$

Which terminal carries the signal fixes the polarity. **Non-inverting** (v_{in} on $+$, V_{REF} on $-$): output high for $v_{in} > V_{REF}$. **Inverting** (v_{in} on $-$, V_{REF} on $+$): output high for $v_{in} < V_{REF}$. Either way the output is binary, and the levels can be made compatible with TTL or any other logic standard.

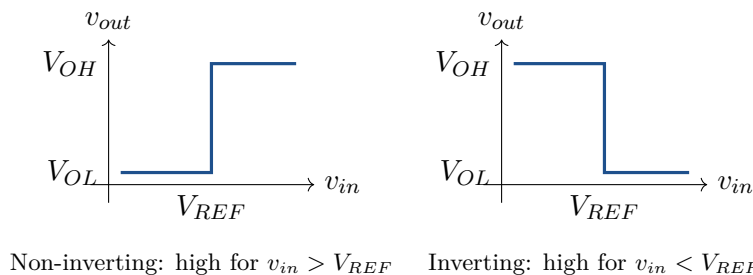


Figure 5.1: Transfer characteristics of the bare comparator: a single step at V_{REF} , with polarity set by which terminal carries the signal.

Watch Out

Near the threshold the bare comparator is *extremely* sensitive to noise (supply ripple, temperature drift, electromagnetic interference). Model the disturbance as a random signal of maximum amplitude $\pm N_A$ riding on v_{in} : the output is guaranteed only for $|v_{in} - V_{REF}| > N_A$. Inside the band $V_{REF} \pm N_A$ the output depends on the instantaneous noise value and shows random oscillations (“chatter”). Hysteresis (next section) is the cure.

Protocol 5.1 Analyse a bare comparator

1. Identify the terminals: which input carries v_{in} , which V_{REF} (generate V_{REF} with a divider if needed).

2. Apply the rule $v_{out} = V_{OH}$ iff $v_d = v_+ - v_- > 0$; sketch the transfer characteristic (one step at V_{REF} , polarity from step 1).
3. Translate to the time domain: the output toggles at every crossing of V_{REF} by v_{in} .
4. **Noise check:** the output is undefined while $|v_{in} - V_{REF}| < N_A$; estimate how long each crossing spends in that band. If chatter is unacceptable, switch to a Schmitt trigger with $\Delta V_S > 2N_A$.

Worked Example 5.1 Squaring-up a triangle wave

A non-inverting comparator with $V_{REF} = 2\text{ V}$ receives a 0–4 V, 1 kHz triangle. Output: a square wave, high while $v_{in} > 2\text{ V}$, i.e. 50% duty cycle. Now let noise of $N_A = 0.1\text{ V}$ ride on the input. The triangle slews at $4\text{ V}/0.5\text{ ms} = 8\text{ V/ms}$, so each crossing spends $2N_A/(8\text{ V/ms}) = 25\text{ }\mu\text{s}$ inside the undefined band 1.9 V–2.1 V — a 25 μs burst of random edges, twice per period. A Schmitt trigger with $\Delta V_S > 0.2\text{ V}$ removes them.

Exercise 5.1

Build an inverting comparator on a single 5 V supply. (a) Choose an E12 divider from the rail so that $V_{REF} = 2.0\text{ V}$. (b) State the output rule. (c) With $N_A = 50\text{ mV}$, give the input band where the output is undefined, and the minimum hysteresis a Schmitt replacement would need.

Solution

(a) $V_{REF} = 5 R_b / (R_a + R_b) = 2\text{ V} \Rightarrow R_a / R_b = 1.5$: take $R_a = 15\text{ k}\Omega$ (top), $R_b = 10\text{ k}\Omega$ (bottom) — both E12, exact. (b) Inverting: $v_{out} = V_{OH}$ for $v_{in} < 2.0\text{ V}$, V_{OL} for $v_{in} > 2.0\text{ V}$. (c) Undefined for $1.95\text{ V} < v_{in} < 2.05\text{ V}$; the design rule $\Delta V_S > 2N_A$ requires at least 0.1 V of hysteresis.

5.2 Inverting Schmitt trigger

Positive feedback makes v_+ depend on v_{out} :

$$v_+ = V_{REF} \frac{R_2}{R_1 + R_2} + v_{out} \frac{R_1}{R_1 + R_2},$$

so the trip point seen by v_{in} ($= v_-$) differs between the two output states. With the output high, the comparator flips when v_{in} rises through V_{S1} ; with it low, it flips back when v_{in} falls through V_{S2} (derivation Appendix D):

$$V_{S1,2} = V_{REF} \frac{R_2}{R_1 + R_2} + V_{OH,OL} \frac{R_1}{R_1 + R_2}, \quad \Delta V_S = V_{S1} - V_{S2} = \frac{R_1}{R_1 + R_2} (V_{OH} - V_{OL}).$$

The *mean* of the hysteresis is

$$V_{S,avg} = \frac{V_{S1} + V_{S2}}{2} = V_{REF} \frac{R_2}{R_1 + R_2} + \frac{V_{OH} + V_{OL}}{2} \frac{R_1}{R_1 + R_2}.$$

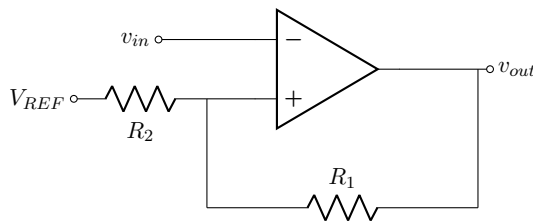
Watch Out

V_{REF} is **not** the mean of the thresholds: $V_{S,avg}$ contains an output-level term $\frac{V_{OH}+V_{OL}}{2} \frac{R_1}{R_1+R_2}$ on top of the attenuated V_{REF} . Even for symmetric levels ($V_{OL} = -V_{OH}$), where that term

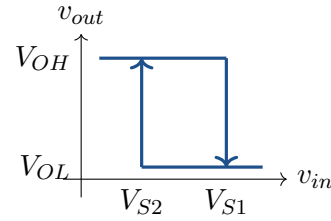
vanishes, $V_{S,avg} = V_{REF} \frac{R_2}{R_1+R_2} \neq V_{REF}$. Centring the hysteresis is a design step, not a freebie.

Protocol 5.2 Design a Schmitt trigger

1. Assume $v_{out} \in \{V_{OH}, V_{OL}\}$ and $i_{\pm} = 0$; write v_+ as the divider of V_{REF} and v_{out} .
2. Solve the two switching points V_{S1} (where a high output flips low) and V_{S2} .
3. From the required $\Delta V_S = V_{S1} - V_{S2}$ get R_2/R_1 ; from one threshold (or from $V_{S,avg}$) get V_{REF} . Choose absolute R_1, R_2 by other constraints.
4. Ensure $\Delta V_S > 2N_A$ (twice the noise amplitude).



Inverting Schmitt trigger



Hysteresis characteristic

Figure 5.2: Positive feedback creates two thresholds; the output is immune to noise smaller than $\Delta V_S = V_{S1} - V_{S2}$.

Worked Example 5.2 Inverting Schmitt

$V_{OL} = 0$, $V_{OH} = 5\text{ V}$, $V_{S1} = 3\text{ V}$, $V_{S2} = 1.5\text{ V}$. $\Delta V_S = 1.5\text{ V} = 5 \cdot \frac{R_1}{R_1+R_2} \Rightarrow R_2 = \frac{7}{3}R_1$. From $V_{S1} = V_{REF} \frac{R_2}{R_1+R_2} + V_{OH} \frac{R_1}{R_1+R_2} = 0.7V_{REF} + 1.5 = 3 \Rightarrow V_{REF} = 2.14\text{ V}$. Pick $R_1 = 10\text{ k}\Omega$, $R_2 = 23.3\text{ k}\Omega$. Check the trap: $V_{S,avg} = \frac{3+1.5}{2} = 2.25\text{ V}$ while $V_{REF} = 2.14\text{ V}$ — the reference is *not* the centre of the hysteresis.

Exercise 5.2

Symmetric Schmitt: $V_{OH} = +12$, $V_{OL} = -12\text{ V}$, thresholds at $\pm 2\text{ V}$. (a) Find V_{REF} and R_2/R_1 and pick E12 values. (b) What is the largest noise amplitude the circuit now tolerates?

Solution

- (a) Symmetric thresholds $\Rightarrow V_{REF} = 0$. Then $\Delta V_S = 4\text{ V} = \frac{R_1}{R_1+R_2} \cdot 24 \Rightarrow \frac{R_1}{R_1+R_2} = \frac{1}{6} \Rightarrow R_2 = 5R_1$: e.g. $R_1 = 10\text{ k}\Omega$, $R_2 = 47\text{ k}\Omega$ (ratio 4.7, $\Delta V_S = 4.2\text{ V}$, thresholds $\pm 2.1\text{ V}$ — fine).
 (b) The rule $\Delta V_S > 2N_A$ gives $N_A < 2.1\text{ V}$: noise up to $\sim 2\text{ V}$ peak cannot cause false switching.

Exercise 5.3 Reverse-engineering a Schmitt

On the bench an inverting Schmitt trigger with $V_{OH} = 10\text{ V}$, $V_{OL} = 0$ is measured to switch at $V_{S1} = 4.0\text{ V}$ and $V_{S2} = 2.8\text{ V}$. (a) Recover R_2/R_1 and V_{REF} . (b) Compare $V_{S,avg}$ with V_{REF} . (c) Propose E12 values and give the thresholds they actually produce.

Solution

(a) $\Delta V_S = 1.2 \text{ V} = 10 \frac{R_1}{R_1 + R_2} \Rightarrow \frac{R_1}{R_1 + R_2} = 0.12 \Rightarrow R_2 = \frac{0.88}{0.12} R_1 = 7.33 R_1$. From $V_{S2} = V_{REF} \cdot 0.88 + V_{OL} \cdot 0.12 = 0.88 V_{REF} = 2.8 \text{ V} \Rightarrow V_{REF} = 3.18 \text{ V}$ (check: $V_{S1} = 0.88 \cdot 3.18 + 1.2 = 4.0 \checkmark$). (b) $V_{S,avg} = 3.4 \text{ V} \neq V_{REF} = 3.18 \text{ V}$: the offset is the V_{OH} pull through the divider. (c) $R_1 = 10 \text{ k}\Omega$, $R_2 = 68 \text{ k}\Omega$ (ratio 6.8): $\frac{R_1}{R_1 + R_2} = 0.128$, so $\Delta V_S = 1.28 \text{ V}$ and, keeping $V_{REF} = 3.18 \text{ V}$, $V_{S1} = 4.05 \text{ V}$, $V_{S2} = 2.77 \text{ V}$ — within 7% of the measured pair.

5.3 Non-inverting Schmitt trigger

Swap the roles: V_{REF} sits on the inverting input and v_{in} reaches the $+$ terminal through R_1 , with R_2 closing the positive-feedback loop (Fig. 5.3). Superposition at the $+$ node gives

$$v_+ = v_{in} \frac{R_2}{R_1 + R_2} + v_{out} \frac{R_1}{R_1 + R_2},$$

and a transition happens when v_+ crosses V_{REF} . Solving for v_{in} once per output state (output low $\rightarrow$ rising threshold V_{S1} ; output high $\rightarrow$ falling threshold V_{S2} ; Appendix D):

$$V_{S1,2} = V_{REF} \frac{R_1 + R_2}{R_2} - V_{OL,OH} \frac{R_1}{R_2}, \quad \Delta V_S = \frac{R_1}{R_2} (V_{OH} - V_{OL}),$$

$$V_{S,avg} = V_{REF} \frac{R_1 + R_2}{R_2} - \frac{V_{OH} + V_{OL}}{2} \frac{R_1}{R_2}.$$

This time the input crosses the threshold in the *same* direction as the output edge: the hysteresis characteristic is traversed counter-clockwise.

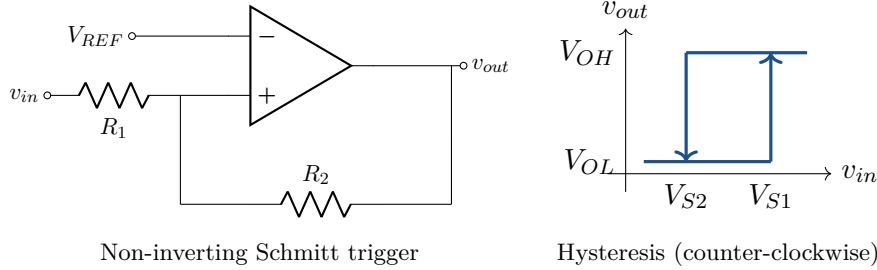


Figure 5.3: Non-inverting Schmitt trigger: the input drives the positive-feedback divider through R_1 ; the comparator trips when v_+ crosses V_{REF} .

Key Idea

Two differences from the inverting topology. First, the hysteresis ratio is R_1/R_2 — not $R_1/(R_1 + R_2)$ — so it can even exceed $V_{OH} - V_{OL}$. Second, the source now *drives* the divider: its output resistance adds to R_1 and widens the hysteresis. Buffer the source, or keep $R_1 \gg R_S$.

Protocol 5.3 Design a non-inverting Schmitt trigger

1. From the required width get the ratio: $R_1/R_2 = \Delta V_S/(V_{OH} - V_{OL})$.
2. From $V_{S,avg}$ (or one threshold) get V_{REF} ; symmetric thresholds with symmetric levels $\Rightarrow V_{REF} = 0$.
3. Pick absolute values: $R_1 \gg$ source resistance, and check the comparator's output

current through R_2 .

4. Verify $\Delta V_S > 2N_A$ and recompute both thresholds with the E12 values actually chosen.

Worked Example 5.3 Same spec, other topology

Repeat the inverting worked example with the non-inverting circuit: $V_{OL} = 0$, $V_{OH} = 5\text{ V}$, $V_{S1} = 3\text{ V}$, $V_{S2} = 1.5\text{ V}$. Width: $\Delta V_S = 1.5\text{ V} = 5 R_1/R_2 \Rightarrow R_2 = \frac{10}{3} R_1$: take $R_1 = 10\text{ k}\Omega$, $R_2 = 33\text{ k}\Omega$ (E12, ratio 3.3 $\Rightarrow \Delta V_S = 1.52\text{ V}$). Reference, from $V_{S1} = V_{REF} \frac{R_1+R_2}{R_2} - V_{OL} \frac{R_1}{R_2} = V_{REF} \cdot \frac{43}{33} = 3\text{ V} \Rightarrow V_{REF} = 2.30\text{ V}$. Check: $V_{S2} = 2.30 \cdot \frac{43}{33} - 5 \cdot \frac{10}{33} = 3.00 - 1.52 = 1.48\text{ V} \checkmark$. Same thresholds as before, but a *different* V_{REF} (2.30 V here vs 2.14 V inverting): the reference never transfers between topologies.

Exercise 5.4

Design a non-inverting Schmitt trigger with $V_{OH,OL} = \pm 12\text{ V}$ and thresholds at $\pm 3\text{ V}$. (a) Find V_{REF} and an E12 pair R_1, R_2 ; give the actual thresholds. (b) The source driving v_{in} has $R_S = 1\text{ k}\Omega$: what do the thresholds become, and what is the lesson?

Solution

(a) Symmetric thresholds and levels $\Rightarrow V_{S,avg} = 0 \Rightarrow V_{REF} = 0$. $\Delta V_S = 6\text{ V} = 24 R_1/R_2 \Rightarrow R_2 = 4R_1$: the E12 pair $R_1 = 8.2\text{ k}\Omega$, $R_2 = 33\text{ k}\Omega$ gives ratio 4.02, $\Delta V_S = 5.96\text{ V}$, thresholds $\pm 2.98\text{ V}$. (b) R_S is in series with R_1 : the effective divider uses $9.2\text{ k}\Omega$, so $\Delta V_S = 24 \cdot 9.2/33 = 6.69\text{ V}$, thresholds $\pm 3.35\text{ V}$ — a +12 % error. Lesson: the non-inverting Schmitt loads its source; buffer it or keep $R_1 \gg R_S$.

5.4 Astable multivibrator (square-wave oscillator)

Feed the inverting Schmitt's output back to its inverting input through an RC (Fig. 5.4). There is no input: the circuit has two *almost*-stable states and oscillates between them forever. While the output is high, the capacitor charges toward V_{OH} along $V_C(t) = V_{OH} + (V_{S2} - V_{OH})e^{-t/\tau}$; when V_C reaches V_{S1} the Schmitt flips, and the capacitor discharges toward V_{OL} until it hits V_{S2} (Fig. 5.5). Solving each exponential for its crossing time ($\tau = RC$; Appendix D):

$$t_1 = \tau \ln \frac{V_{S2} - V_{OH}}{V_{S1} - V_{OH}}, \quad t_2 = \tau \ln \frac{V_{S1} - V_{OL}}{V_{S2} - V_{OL}}, \quad T = t_1 + t_2 = \tau \ln \frac{(V_{S2} - V_{OH})(V_{S1} - V_{OL})}{(V_{S1} - V_{OH})(V_{S2} - V_{OL})}.$$

The two half-periods need not be equal: the duty cycle $\delta = t_1/T$ is 50 % only when thresholds and levels are symmetric.

Key Idea

Symmetric supplies and thresholds ($V_{OL} = -V_{OH}$, $V_{REF} = 0$, so $V_{S1,2} = \pm V_{OH} \frac{R_1}{R_1+R_2}$) collapse the period to the classic

$$T = 2\tau \ln \left(1 + \frac{2R_1}{R_2} \right),$$

with equal half-periods and exactly 50 % duty cycle. With $R_1 = R_2$: $T = 2\tau \ln 3 \approx 2.2\tau$.

Protocol 5.4 Design an astable oscillator

1. From the required T (or $f = 1/T$) and the threshold levels, solve for $\tau = RC$.
2. Choose R and C (one degree of freedom) to give that τ .
3. From ΔV_S get R_2/R_1 (as in the Schmitt protocol).
4. Check the OPAMP output current into the feedback network.

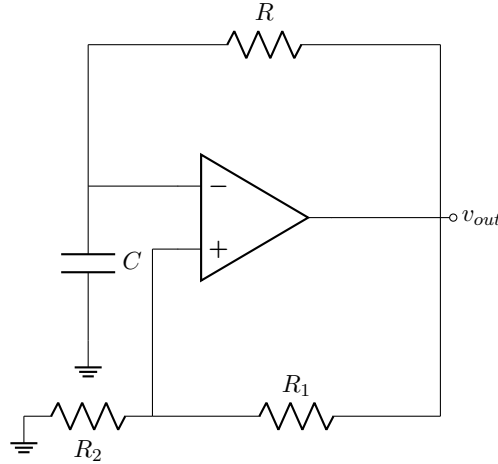


Figure 5.4: Astable multivibrator: R, C set the timing; R_1, R_2 set the thresholds via positive feedback.

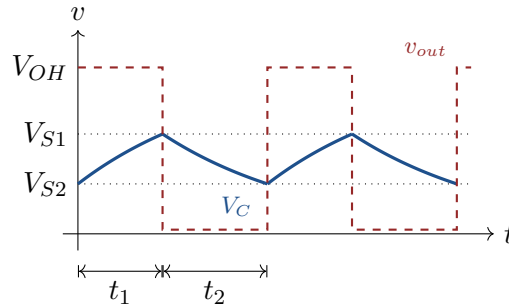


Figure 5.5: Astable timing for the 50 kHz worked example ($V_{OH} = 5\text{ V}$, $V_{OL} = 0$): the capacitor (solid) rides two exponentials between V_{S2} and V_{S1} while the output (dashed) squares. Because levels and thresholds are not symmetric, $t_1 \neq t_2$.

Worked Example 5.4 50 kHz oscillator

$T = 20\text{ }\mu\text{s}$, $V_{OH} = 5\text{ V}$, $V_{OL} = 0$, $V_{S1} = 3\text{ V}$, $V_{S2} = 1.5\text{ V}$. The period formula gives $T = \tau \cdot 1.253 \Rightarrow \tau = 16\text{ }\mu\text{s}$. Choose $R = 20\text{ k}\Omega \Rightarrow C = 800\text{ pF}$. From ΔV_S : $R_2/R_1 = 7/3$, e.g. $R_1 = 10\text{ k}\Omega$, $R_2 = 23.3\text{ k}\Omega$.

Note

The astable is a generator, not a conditioning block: it has no input, and at power-up it falls into the loop after one short transient. With $V_{REF} = 0$ it needs a *bipolar* supply; on a single supply add a V_{REF} divider so both thresholds sit inside the charging range. Finally check I_{MAX} : the output drives the timing resistor R and the $R_1 + R_2$ divider in parallel

with the load.

Exercise 5.5 Duty cycle of the 50 kHz oscillator

For the worked example above ($V_{OH} = 5\text{ V}$, $V_{OL} = 0$, $V_{S1} = 3\text{ V}$, $V_{S2} = 1.5\text{ V}$): (a) compute t_1 , t_2 and the duty cycle; (b) explain why the symmetric design of the next exercise gives exactly 50%.

Solution

(a) $t_1 = \tau \ln \frac{1.5-5}{3-5} = \tau \ln 1.75 = 0.560 \tau$ (output high), $t_2 = \tau \ln \frac{3-0}{1.5-0} = \tau \ln 2 = 0.693 \tau$; the sum confirms $T = 1.253 \tau$. Duty cycle $\delta = 0.560/1.253 = 44.7\%$ — visibly asymmetric, exactly as expected. (b) With $V_{OL} = -V_{OH}$ and $V_{S1,2} = \pm k V_{OH}$ both logarithm arguments equal $\frac{1+k}{1-k}$, so $t_1 = t_2$ and $\delta = 50\%$.

Exercise 5.6

Design a 10 kHz astable with $V_{OH} = +12$, $V_{OL} = -12\text{ V}$ and symmetric thresholds at half the output levels, $V_{S1,2} = \pm 6\text{ V}$.

Solution

Thresholds: $\Delta V_S = 12\text{ V} = \frac{R_1}{R_1+R_2} \cdot 24 \Rightarrow R_1 = R_2$, e.g. both $10\text{ k}\Omega$. Period: the log argument is $\frac{(V_{S2}-V_{OH})(V_{S1}-V_{OL})}{(V_{S1}-V_{OH})(V_{S2}-V_{OL})} = \frac{(-18)(18)}{(-6)(6)} = 9$, so $T = \tau \ln 9 = 2.20 \tau$; $T = 100\text{ }\mu\text{s} \Rightarrow \tau = 45.5\text{ }\mu\text{s}$: $C = 1\text{ nF}$, $R = 45.5\text{ k}\Omega$ (E12 $47\text{ k}\Omega$ gives $f = 9.7\text{ kHz}$, -3%). This is the classic symmetric result $T = 2\tau \ln 3$.

5.5 Triangular + square wave generator

The astable's capacitor wave *looks* triangular but is a chain of exponentials — fine for timing, poor as a test signal. For true straight ramps, replace the passive RC with the **integrator** of §4.3: a constant input then gives an exactly constant slope. The integrator inverts, so the comparator must be the *non-inverting* Schmitt of §5.3 (with $V_{REF} = 0$), or the loop would latch instead of oscillating. The square wave v_{sq} drives the integrator, whose output v_{tri} ramps at slope $-v_{sq}/RC$ and is also the comparator's input, closing the loop (Fig. 5.7).

The triangle's peaks *are* the Schmitt thresholds: the comparator trips when its $+$ node crosses zero, i.e. when the triangle reaches $\mp v_{sq} R_1/R_2$, so for symmetric levels the thresholds are $\pm V_{OH} R_1/R_2$. Each ramp crosses ΔV_S at constant speed $|v_{sq}|/RC$, giving

$$V_{S1} - V_{S2} = (V_{OH} - V_{OL}) \frac{R_1}{R_2}, \quad t_1 = \frac{RC (V_{S1} - V_{S2})}{-V_{OL}}, \quad t_2 = \frac{RC (V_{S1} - V_{S2})}{V_{OH}},$$

$$T = t_1 + t_2 \xrightarrow{V_{OL}=-V_{OH}} 2RC \frac{V_{S1} - V_{S2}}{V_{OH}}.$$

Because the integrator sees a square wave that averages to zero, it never winds up and needs no periodic reset — though the stabilising parallel resistance of §4.3 is still good practice.

Protocol 5.5 Design a triangle/square generator

1. **Amplitude:** the triangle's peaks *are* the Schmitt thresholds; set R_1/R_2 from $V_{S1} - V_{S2} = (V_{OH} - V_{OL}) R_1/R_2$.

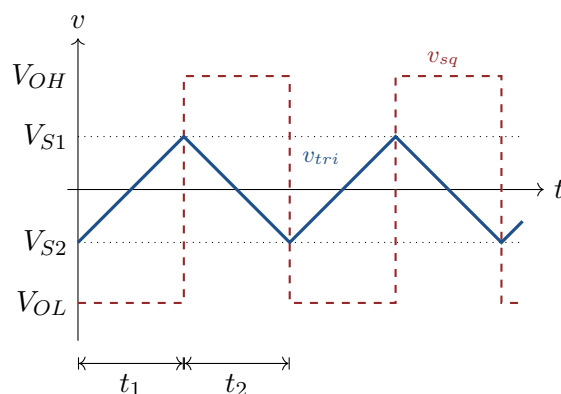


Figure 5.6: Generator waveforms (symmetric levels): the triangle *rises* while the square sits at V_{OL} (the integrator inverts) and turns exactly at the Schmitt thresholds $\pm V_{OH}R_1/R_2$.

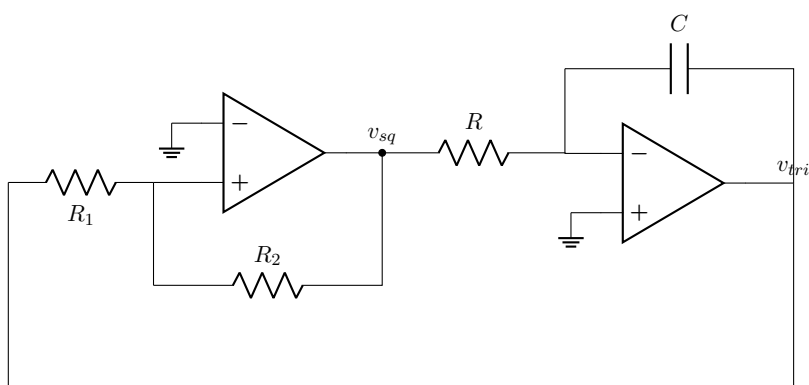


Figure 5.7: Triangle/square generator: the Schmitt's square output v_{sq} is integrated into the triangle v_{tri} , which returns through R_1 and trips the comparator at $\pm V_{OH}R_1/R_2$.

2. **Period:** the integrator ramps between the thresholds at slope V_{OH}/RC , so $T = 2RC(V_{S1} - V_{S2})/V_{OH}$ fixes RC ; split it into R and C .
3. **Offset (if unipolar):** to shift the triangle, put a V_{REF} on the comparator's $-$ terminal and recompute the thresholds with the non-inverting Schmitt formulas of §5.3.
4. **Check the parts:** the comparator must slew fast enough for clean square edges, and its I_{MAX} must feed both R_2 and R .

Worked Example 5.5 500 Hz, 8 V triangle

$V_{supply} = \pm 15\text{ V}$, $V_{OH} = 12\text{ V}$, $V_{OL} = -12\text{ V}$. Amplitude: $V_{S1} - V_{S2} = 8\text{ V} = 24 \cdot R_1/R_2 \Rightarrow R_2 = 3R_1$, e.g. $R_1 = 47\text{ k}\Omega$, $R_2 = 150\text{ k}\Omega$. Period: $T = 2\text{ ms} = 2RC \cdot 8/12 \Rightarrow RC = 1.5\text{ ms}$, e.g. $R = 100\text{ k}\Omega$, $C = 15\text{ nF}$.

Note

Three menu variants, one knob each. *Variable frequency:* the period is linear in RC , so make R a potentiometer. *Variable offset:* insert a V_{REF} on the comparator's $-$ terminal (potentiometer-driven) to shift the triangle — see the unipolar exercise below. *Variable duty cycle:* split R into two diode-selected resistances, one per ramp direction; unequal values

turn the triangle into a sawtooth.

Exercise 5.7

Design for a 1 kHz, 5 V peak-to-peak triangle with $V_{OH}, V_{OL} = \pm 10$ V.

Solution

Amplitude: $5 = 20 R_1/R_2 \Rightarrow R_2 = 4R_1$: $R_1 = 10$ k Ω , $R_2 = 39$ k Ω (E12, ratio 3.9, $V_{pp} = 5.1$ V). **Period:** $T = 1$ ms $= 2RC \cdot 5/10 \Rightarrow RC = 1$ ms: $C = 10$ nF, $R = 100$ k Ω .

Exercise 5.8 Unipolar generator

From ± 15 V supplies ($V_{OH,OL} = \pm 12$ V), generate a *unipolar* 0–5 V, 1 kHz triangle by adding a V_{REF} to the comparator's $-$ terminal. Find R_1, R_2 (E12), V_{REF} , the actual thresholds, and R, C .

Solution

Amplitude: $\Delta V_S = 5$ V $= 24 R_1/R_2 \Rightarrow R_2 = 4.8 R_1$: take $R_1 = 10$ k Ω , $R_2 = 47$ k Ω ($\Delta V_S = 24 \cdot 10/47 = 5.11$ V). **Offset:** the comparator now trips when $v_+ = V_{REF}$, so the non-inverting Schmitt formulas of §5.3 apply. Symmetric levels kill the $V_{OH} + V_{OL}$ term: $V_{S,avg} = V_{REF} \frac{R_1+R_2}{R_2} = V_{REF} \cdot \frac{57}{47} = 2.5$ V $\Rightarrow V_{REF} = 2.06$ V. Check: $V_{S1} = 2.06 \cdot \frac{57}{47} + 12 \cdot \frac{10}{47} = 2.50 + 2.55 = 5.05$ V, $V_{S2} = 2.50 - 2.55 = -0.05$ V — 0–5 V within the E12 rounding. **Period:** $T = 1$ ms $= 2RC \cdot 5.11/12 \Rightarrow RC = 1.17$ ms: $C = 10$ nF, $R = 120$ k Ω (E12; $f = 980$ Hz, -2%). Generate V_{REF} from the $+15$ V rail with a stiff divider or a buffered potentiometer.

Chapter 6

Laboratory: Op-amp characterisation

This lab measures, on a real LM741-class part, the quantities the earlier chapters predict: closed-loop gain (Ch. 1), input/output resistance and output swing (Ch. 2), and the frequency response (Ch. 3). A function generator drives the input, a DSO reads input and output, and a $\pm 15\text{ V}$ supply powers the rails.

Key Idea

The theory chapters give the *expected* value; the lab is where you confront it with a measurement and its uncertainty. The recurring trick is the *difference method*: toggle one element (a series resistor, a load) and read how a node moves — the change reveals R_{in} , R_{out} or a current limit.

6.1 Inverting amplifier: gain

Protocol 6.1 Measure closed-loop gain

1. Apply a small sinusoid (e.g. 1 kHz, $V_{pp} \sim 1\text{ V}$) well inside the linear range.
2. Read $V_{pp,in}$ and $V_{pp,out}$; the measured gain is $A_V = -V_{pp,out}/V_{pp,in}$.
3. Compare with the design value $A_V = -R_f/R_{in}$.

With $R_f = 100\text{ k}\Omega$, $R_{in} = 22\text{ k}\Omega$ the design gain is $-100/22 = -4.545$; the bench gave $-4.77/1.06 = -4.54$ — agreement to a fraction of a percent. Adding a 500 mV DC offset leaves the AC gain unchanged: the amplifier processes the AC component independently of the DC operating point.

6.2 Step response: bandwidth- vs slew-rate-limited

Driving a 1 kHz square and reading the output rise time separates two regimes (Ch. 3). For a *small* step ($V_{pp} = 0.1\text{ V}$ in, $t_{rise} \approx 1.4\text{ }\mu\text{s}$) the edge is *bandwidth-limited* ($t_{rise} \approx 0.35/\text{BW}$). For a *large* step ($V_{pp} = 2\text{ V}$ in, $t_{rise} \approx 9\text{ }\mu\text{s}$) the slope hits the *slew rate* and the edge becomes a straight ramp.

Watch Out

A large-signal rise time does *not* give the bandwidth — it gives the slew rate, $\text{SR} \approx \Delta V/\Delta t$. Only small-signal edges obey $t_{rise} = 0.35/\text{BW}$.

6.3 Virtual ground

Probing both inputs with V^+ grounded: V^- sits within millivolts of zero (here ~ 0.5 mV average, the rest noise). This is the *virtual ground* — negative feedback drives $v_d = v_{out}/A_d$ toward zero, so with $A_d \approx 2 \times 10^5$ the residual difference is $\sim 10^5$ times smaller than V_{out} and 180° out of phase with it.

6.4 Output dynamic range and current limit

Loading the output exposes the op-amp's current ceiling (it is not a power driver).

Worked Example 6.1 Driving into the current limit

With $R_{11} = 1$ k Ω the output draws $I \approx 4.8$ mA (under the ~ 25 mA typical limit) and the sinusoid is clean. Adding $R_{12} = 100$ Ω in parallel demands $I \approx 48$ mA — above the limit — so the output *clips*: V_{min} rises off its expected value, the average shifts away from zero, and the wave distorts. The op-amp has entered current limiting/saturation.

6.5 Non-inverting amplifier: R_{in} and R_{out}

Both use the difference method. **Input resistance:** switching a series R_3 in/out forms a divider with the source node; from the two readings $R_{seen} = R_3 V_{open}/(V_{closed} - V_{open})$, then undo the bias resistor to get $R_{in} \approx 2.1$ M Ω . **Output resistance:** inserting a load and reading the output drop gives the *closed-loop* $R_{out,CL} \approx 18$ Ω , lower than the open-loop ~ 75 Ω by the loop gain, $R_{out,CL} = R_{out,OL}/(1 + A_d\beta)$.

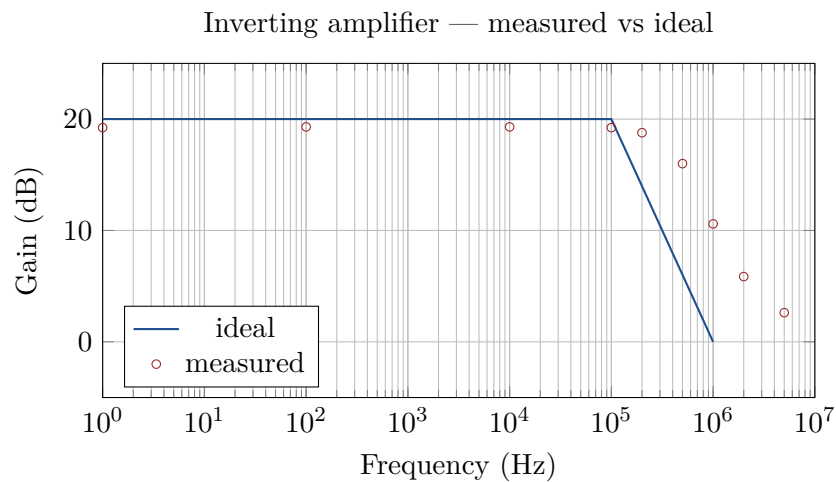
Watch Out

Undoing the input divider subtracts two nearly equal numbers ($R_{seen} \approx R_{bias}$): the relative uncertainty explodes (Ch. 2). Quote R_{in} as an order of magnitude, not to three digits.

6.6 Frequency response (Bode)

Protocol 6.2 Measure a Bode magnitude plot

1. Hold the input amplitude constant; sweep the frequency in decade steps.
2. At each point read $A_V = V_{out}/V_{in}$ and convert to $20 \log_{10} A_V$ (in dB).
3. Plot dB versus $\log f$; the -3 dB corner is the closed-loop bandwidth, beyond which the gain rolls off at 20 dB per decade.



The midband gain (≈ 19.3 dB) matches $20 \log_{10}(R_f/R_{in})$; the measured corner and roll-off track the gain–bandwidth product (Ch. 3).

6.7 Audio amplifier

The audio stage is deliberately *band-limited*: a high-pass at the low end and the GBWP roll-off at the high end give a band-pass magnitude response, flat at ≈ 19 dB across roughly 1 kHz–20 kHz and falling on both sides (Ch. 4, audio design).

Appendix A

Ideal rules and feedback

With $A_d \rightarrow \infty$ a finite output forces $v_d = v_{out}/A_d \rightarrow 0$, hence $v_+ \simeq v_-$; with $r_d \rightarrow \infty$ the input currents vanish. These are the two golden rules used throughout Chapter 1. Each table entry follows by KCL/KVL at the (often virtual-ground) input node.

A.1 Feedback factor and loop gain

For the standard topology $\beta = R_1/(R_1 + R_2)$ and $T = A_d\beta$. Writing $Y = A_d(I - \beta Y)$ gives the closed-loop gain $Y/I = \frac{1}{\beta} \frac{T}{1+T}$, which $\rightarrow 1/\beta$ as $T \rightarrow \infty$. Feedback also scales impedances by $(1 + T)$.

Appendix B

Real-OpAmp derivations

Finite A_d . From $v_{in} - v_{out}/A_d = \beta v_{out}$, $A_V = v_{out}/v_{in} = \frac{1}{\beta} \frac{T}{1+T}$.

B.1 Test-source R_{in}, R_{out}

Applying V_x at the input and using $v_{out} = A_d r_d I_x$ in the feedback mesh yields $V_x = I_x [r_d(1 + A_d\beta) + \beta R_2] \approx I_x r_d(1 + T)$, so $R_{in} = r_d(1 + T)$. At the output, killing the input and applying V_x gives $R_{out} = r_o/(1 + T)$.

B.2 CMRR and PSRR

Write the differential amplifier's output as $v_{out} = A_1 V_1 + A_2 V_2$ and change variables to $v_d = V_1 - V_2$, $V_C = \frac{1}{2}(V_1 + V_2)$. Matching coefficients gives $v_{out} = A_d v_d + A_C V_C$ with $A_d = (A_1 - A_2)/2$ and $A_C = A_1 + A_2$, hence $\text{CMRR} \triangleq A_d/A_C$ and $\text{CMRR}_{dB} = 20 \log_{10}(A_d/A_C)$.

Carrying the extra $A_C V_C$ term through the non-inverting feedback analysis (with $V_1 = v_{in}$ and $V_2 = \beta v_{out}$, so $V_C = \frac{1}{2}(v_{in} + \beta v_{out})$) and solving for v_{out}/v_{in} gives the *exact* closed-loop gain

$$\frac{v_{out}}{v_{in}} = \frac{1}{\beta} \frac{A_d \beta \left(1 + \frac{1}{2 \text{CMRR}}\right)}{1 + A_d \beta \left(1 - \frac{1}{2 \text{CMRR}}\right)}.$$

For any usable device $\text{CMRR} \gg 1$, the $1/(2 \text{CMRR})$ corrections vanish and this collapses to the familiar $\frac{1}{\beta} \frac{T}{1+T}$ — so a real, non-zero A_C is harmless once feedback is closed. The **PSRR** (Power-Supply Rejection Ratio) is defined and applied identically, but measures the output's sensitivity to a *supply-voltage* change rather than to a common-mode input; datasheets tabulate both (for the LM741, CMR and SVR are each ≈ 86 dB).

B.3 Offset by superposition

Treat V_{OFF} , I_+ , I_- one at a time. V_{OFF} sees a non-inverting gain $1 + R_2/R_1$; I_- flows in R_2 ; I_+ drops across R_3 and is amplified. Summing and substituting $I_{\pm} = I_{BIAS} \pm I_{OFF}/2$ gives the boxed result in Section 2.6; $R_3 = R_1 \parallel R_2$ nulls the I_{BIAS} term.

Appendix C

Frequency response derivations

Each internal pole adds -20 dB/dec and -90° . The feedback is negative while the total shift $\phi_{FB} = 180^\circ + \angle T < 360^\circ$; positive feedback (oscillation risk) appears where $\angle T \rightarrow -180^\circ$. The audio amplifier's three operating regimes (DC / pass band / ∞) follow by shorting or opening each capacitor.

C.1 Gain–Bandwidth Product

With one dominant pole, $A_d(f) = A_{d0}/(1 + jf/f_0)$ and $A_V(f) \approx \frac{1}{\beta} \frac{1}{1 + jf/(A_{d0}\beta f_0)}$, a single pole at $f_T = A_{d0}f_0\beta = f_{BW}/A_{V0}$. Hence $\text{gain} \times \text{bandwidth} = f_{BW}$ (constant) for an internally compensated part.

Appendix D

Oscillator and comparator timing

Inverting Schmitt: $v_+ = V_{REF} \frac{R_2}{R_1+R_2} + v_{out} \frac{R_1}{R_1+R_2}$; setting $v_- = v_{in} = v_+$ at each output level gives $V_{S1,2} = V_{REF} \frac{R_2}{R_1+R_2} + V_{OH,OL} \frac{R_1}{R_1+R_2}$, hence $\Delta V_S = \frac{R_1}{R_1+R_2} (V_{OH} - V_{OL})$ and $V_{S,avg} = V_{REF} \frac{R_2}{R_1+R_2} + \frac{V_{OH}+V_{OL}}{2} \frac{R_1}{R_1+R_2}$.

Non-inverting Schmitt: now $v_+ = v_{in} \frac{R_2}{R_1+R_2} + v_{out} \frac{R_1}{R_1+R_2}$ and the trip condition is $v_+ = V_{REF}$; solving for v_{in} at each output level gives $V_{S1,2} = V_{REF} \frac{R_1+R_2}{R_2} - V_{OL,OH} \frac{R_1}{R_2}$, hence $\Delta V_S = \frac{R_1}{R_2} (V_{OH} - V_{OL})$ and $V_{S,avg} = V_{REF} \frac{R_1+R_2}{R_2} - \frac{V_{OH}+V_{OL}}{2} \frac{R_1}{R_2}$.

Astable: each half-period solves the RC charge equation $V_C(t) = V_\infty + (V_0 - V_\infty)e^{-t/\tau}$, with $(V_0, V_\infty) = (V_{S2}, V_{OH})$ while rising and (V_{S1}, V_{OL}) while falling: $t_1 = \tau \ln \frac{V_{S2}-V_{OH}}{V_{S1}-V_{OH}}$, $t_2 = \tau \ln \frac{V_{S1}-V_{OL}}{V_{S2}-V_{OL}}$, $T = t_1 + t_2$.

Triangle/square generator: the integrator output ramps linearly, $v_{tri}(t) = v_{tri}(0) - \frac{v_{sq}}{RC} t$; equating each ramp's excursion to $V_{S1} - V_{S2}$ gives $t_1 = RC \frac{V_{S1}-V_{S2}}{-V_{OL}}$ (square low, triangle rising), $t_2 = RC \frac{V_{S1}-V_{S2}}{V_{OH}}$, and the period of Chapter 5.

Appendix E

Reference data

E12 series (5% tolerance), all decades: 1.0, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2. *Rule:* the relative error of a product or ratio of quantities is the sum of their relative errors; an additive constant (the “1” in $1 + R_2/R_1$) carries no error.

Table E.1: Representative OPAMP parameters used in the examples.

Parameter	LM741	TL082
DC gain A_d0	$\sim 200\text{ V/mV}$	$\sim 200\text{ V/mV}$
CMRR	86 dB	86 dB
GBWP	1 MHz	4 MHz
Slew rate	$0.5\text{ V}/\mu\text{s}$	$13\text{ V}/\mu\text{s}$

Appendix F

Glossary of core terms

The protocols and worked examples assume you already know what each word means. This glossary supplies the missing one-line explanations: every core term, grouped by where it first appears, in plain language with its defining relation and a link to the section that uses it. It closes with a symbol key (Table F.1).

F.1 The device and its ideal model

Operational amplifier (OpAmp). A high-gain differential voltage amplifier with two inputs and one output; internally a differential pair, a gain stage and a power output stage. Its job is to amplify the *difference* of its inputs: $v_{out} = A_d(v_+ - v_-)$ (Fig. 1.1).

Non-inverting input (+) / inverting input (-). The two input terminals. A signal entering + reaches the output with the *same* sign; one entering - is *inverted* (180° phase shift).

Differential input voltage $v_d = v_+ - v_-$. The voltage actually amplified. Negative feedback drives it toward zero, which is the basis of the virtual short.

Differential (open-loop) gain A_d . The factor multiplying v_d in $v_{out} = A_d v_d$ when there is *no* feedback. It is huge (10^4 – 10^6) but a *technological* parameter — known only to poor accuracy and varying with part, temperature and frequency. The whole design philosophy is therefore: make the result depend on resistor *ratios*, not on A_d .

Power-supply rails V_{cc}/V_{dd} (**positive**), V_{ee}/V_{ss} (**negative**). The supply terminals (the negative one may be ground). Usually omitted from schematics but always present in hardware; the output can never swing beyond them.

Linear region vs. saturation. While $|v_d|$ is tiny the output follows $v_{out} = A_d v_d$ (*linear region*). Once that product would exceed a rail the output sticks just short of it (*saturation*, $V_{OH} \approx V_{cc}$, $V_{OL} \approx V_{ee}$). The linear window is vanishingly narrow, $\Delta v_{d,lin} = (V_{cc} - V_{ee})/A_d$ — only $300\text{ }\mu\text{V}$ for $\pm 15\text{ V}$ and $A_d = 10^5$ — which is exactly why a bare OPAMP is useless as an amplifier and needs feedback.

Rail-to-rail. An OPAMP whose output (and sometimes input) range extends all the way to the supply rails, rather than stopping a volt or so short.

Virtual short / virtual ground. Consequences of negative feedback with $A_d \rightarrow \infty$: since v_{out} is finite, $v_d = v_{out}/A_d \rightarrow 0$, so $v_+ \simeq v_-$ (*virtual short*). If the + terminal is grounded then $v_- = 0$ even though no current flows there (*virtual ground*).

The two golden rules. The pair that makes ideal-OPAMP analysis mechanical: (1) $v_+ \simeq v_-$ (virtual short); (2) $i_+ = i_- = 0$ (no input current). They follow from $A_d \rightarrow \infty$ and $r_d \rightarrow \infty$ (Appendix A).

F.2 The five amplifier building blocks

Non-inverting / inverting voltage amplifier. Voltage-in, voltage-out ($V \rightarrow V$): $A_V = 1 + R_2/R_1$ (always ≥ 1) or $A_V = -R_2/R_1$ respectively (Table 1.1).

Transresistance amplifier. Current-in, voltage-out ($I \rightarrow V$). Its gain $A_R = -R_2$ has units of *ohms*; the $-$ node is a virtual ground so $v_{out} = -R_2 i_{in}$.

Transconductance amplifier. Voltage-in, current-out ($V \rightarrow I$). Its gain $g_m = 1/R_1$ has units of *siemens* (a conductance), so $i_{out} = v_{in}/R_1$. *Note:* this gain is sometimes labelled A_C (“C” for conductance); do not confuse it with the common-mode gain A_C of the real device.

Current amplifier. Current-in, current-out ($I \rightarrow I$): $i_{out} = (1 + R_2/R_1) i_{in}$, independent of the load.

Voltage follower (buffer). The non-inverting amplifier at unity gain ($R_1 = \infty$, $R_2 = 0$, so $\beta = 1$): $v_{out} = v_{in}$. With $\beta = 1$ it has the best possible impedances (highest R_{in} , lowest R_{out}), at the price of no amplification (Fig. 1.2c).

Impedance matching. The follower’s job: it lets a source with output resistance R_S deliver its *full* voltage to a load R_L , defeating the $R_L/(R_S + R_L)$ divider loss that a direct connection would suffer.

F.3 Feedback

Negative feedback. Returning part of the output to the $-$ input. It makes the closed-loop gain depend on resistors instead of A_d , multiplies R_{in} and divides R_{out} by $(1 + T)$, and widens the bandwidth. Almost every linear OPAMP circuit uses it.

Positive feedback. Returning part of the output to the $+$ input. It is *regenerative* — it pushes the output toward a rail — and is the basis of the Schmitt trigger and the oscillators (Chapter 5).

Feedback factor β . The fraction of the output fed back, $\beta = R_1/(R_1 + R_2)$ for the standard topology. The ideal closed-loop gain is simply $1/\beta$ (Appendix A.1).

Loop gain $T = A_d\beta$. The gain going once around the whole loop. It is the single number that says “how ideal” the circuit is: the closed-loop gain is $G = \frac{1}{\beta} \frac{T}{1+T}$, which tends to $1/\beta$ as $T \rightarrow \infty$. A *large* T means feedback works well; a high-gain stage (small β) has a small T and reaps fewer of feedback’s benefits.

Open-loop vs. closed-loop gain. Open-loop is the raw A_d (no feedback); closed-loop (A_V , G) is the controlled gain once feedback is wrapped around it.

F.4 Real (non-ideal) parameters

Finite A_d . Makes the gain fall *below* the ideal value by the factor $T/(1 + T)$. Usually a negligible error (a fraction of a percent), far smaller than resistor tolerances (Appendix B).

Input resistance r_d / output resistance r_o of the OpAmp. The device’s own (open-loop) port resistances. Feedback transforms them into the *amplifier’s* $R_{in} = r_d(1 + T)$ (very large) and $R_{out} = r_o/(1 + T)$ (very small), found by the test-source method (Appendix B.1).

Maximum output current I_{MAX} . The largest current the output can source or sink (often ~ 20 mA). It must feed both the load and the feedback network, so the feedback share is kept small ($I_{fb} < I_{load}/10$) — this sets a *lower* bound on R_2 .

Common mode $V_C = \frac{1}{2}(V_1 + V_2)$. The part of the input common to both terminals (their

average). An ideal OPAMP ignores it; a real one amplifies it slightly.

Common-mode gain A_C . The unwanted gain applied to V_C (ideally 0). It measures the amplifier's input *asymmetry*: writing $v_{out} = A_1 V_1 + A_2 V_2$, one has $A_C = A_1 + A_2$ (zero only when $A_1 = -A_2$ exactly).

CMRR (common-mode rejection ratio). $CMRR = A_d/A_C$, usually quoted in dB as $20 \log_{10}(A_d/A_C)$. The higher, the better the OPAMP rejects what is common to both inputs (mains hum, ground shifts); feedback suppresses the common-mode contribution by a further factor of ~ 2 CMRR (Appendix B.2).

PSRR (power-supply rejection ratio). The supply-side twin of CMRR: the same $20 \log_{10}$ ratio, but measuring how little the output moves when the *supply voltage* changes rather than when a common-mode input is applied. Datasheets list it next to CMRR (for the LM741 both ≈ 86 dB).

Input offset voltage V_{OFF} . A small spurious differential voltage (millivolts) that behaves as if applied between the inputs; with zero real input it still produces an amplified output. It cannot be cancelled by adding resistors — only by a better part or a trim (§2.6).

Input bias current I_{BIAS} . The average DC current the two inputs draw to bias their internal transistors. Its contribution to the output offset is cancelled by an extra resistor $R_3 = R_1 \parallel R_2$ on the + terminal.

Input offset current I_{OFF} . The *imbalance* between the two input bias currents (smaller than I_{BIAS}). What remains after R_3 cancels the bias term; it is bounded by keeping R_2 small enough (Appendix B.3).

Output offset voltage. The total non-zero output seen with zero input, summing the V_{OFF} , I_{BIAS} and I_{OFF} contributions by superposition.

Dynamic range. The allowed span of a quantity. The *input differential*, *input common-mode* and *output* ranges are each bounded by the supplies (and the output additionally by $I_{MAX} R_L$); rail-to-rail parts maximise them.

F.5 Frequency response and dynamics

Pole / Bode plot. A_d is not constant with frequency: each internal *pole* f_p bends its asymptotic magnitude down by 20 dB/decade and adds -90° of phase. The Bode plot is the magnitude/phase-vs-frequency picture used to judge stability (Fig. 3.1, Appendix C).

Bandwidth f_{BW} . The frequency at which the closed-loop gain has fallen by 3 dB; the usable small-signal frequency span of the amplifier.

Gain–Bandwidth Product (GBWP, f_T). For an internally compensated OPAMP the product gain $\times$ bandwidth is constant: $f_{BW} = GBWP/A_{V0}$. More gain therefore costs proportionally less bandwidth (Appendix C.1).

Phase margin ϕ_M . $\phi_M = 180^\circ + \angle T$ measured where $|T| = 0$ dB. It is the safety margin before negative feedback turns into positive feedback; a design wants $\phi_M \geq 45^\circ$.

Stability / compensation. Stable means the fed-back amplifier does not oscillate. *Compensation* reshapes the poles (typically by lowering the first pole) so $|T|$ crosses 0 dB before the phase reaches -180° .

Internally compensated OpAmp. A part with a built-in dominant pole, guaranteed stable for any closed-loop gain ≥ 1 (the easy, “just use it” choice).

Slew rate (SR). The maximum rate of change of the output, $SR = I_s/C_c$, set by how fast the compensation capacitor can be charged. It is a *large-signal* limit (distinct from bandwidth): a

sine of amplitude V_p avoids slewing only while $V_p \leq SR/(2\pi f)$ (the full-power bandwidth).

F.6 Comparators and oscillators

Comparator. An OPAMP used *without* negative feedback: v_d is no longer driven to zero, so the output saturates to V_{OH} when $v_+ > v_-$ and to V_{OL} otherwise. It reports which input is larger.

Output levels V_{OH}, V_{OL} . The high and low saturated output voltages of a comparator (near the rails, or 0/5 V for a single-supply logic-level part).

Hysteresis. Two distinct switching thresholds, $V_{S1} > V_{S2}$, created by positive feedback. Between them the output “remembers” its last state, so noise smaller than $\Delta V_S = V_{S1} - V_{S2}$ cannot cause false chatter.

Noise amplitude N_A . Maximum amplitude of the random disturbance riding on v_{in} ; a bare comparator’s output is undefined while $|v_{in} - V_{REF}| < N_A$ (§5.1).

Schmitt trigger. A comparator with hysteresis. Design rule: make $\Delta V_S > 2N_A$ (twice the noise amplitude) (Fig. 5.2, Protocols 5.2–5.3).

Threshold voltages V_{S1}, V_{S2} and width ΔV_S . The two trip points and their separation; $\Delta V_S = \frac{R_1}{R_1+R_2}(V_{OH} - V_{OL})$ for the inverting topology, $\frac{R_1}{R_2}(V_{OH} - V_{OL})$ for the non-inverting one (Appendix D).

Hysteresis mean $V_{S,avg}$. $\frac{1}{2}(V_{S1} + V_{S2})$, the centre of the hysteresis window. It is *not* equal to V_{REF} : an output-level term is added through the divider (§5.2).

Astable multivibrator. A square-wave oscillator: a Schmitt trigger whose output charges/discharges an RC between its two thresholds, with period set by $\tau = RC$ and the levels.

Integrator. An inverting stage whose feedback resistor is replaced by a capacitor: $v_{out}(t) = -\frac{1}{RC} \int v_{in} dt$ (a pole at the origin). A constant input yields a ramp; a square wave yields a triangle (Fig. 4.2).

F.7 Symbol and notation quick-reference

Table F.1: Symbols used throughout the manual.

Symbol	Meaning	Defining relation / typical
A_d	open-loop differential gain	$v_{out} = A_d v_d$; 10^4 – 10^6
A_{d0}	DC (low-frequency) value of A_d	e.g. 100 dB
$v_d = v_+ - v_-$	differential input voltage	$\rightarrow 0$ with feedback
v_+, v_-	non-inverting / inverting input voltages	$v_+ \simeq v_-$ (virtual short)
A_V	closed-loop voltage gain	$1 + R_2/R_1$ or $-R_2/R_1$
A_R	transresistance gain	$-R_2$ (units Ω)
g_m	transconductance gain (also written A_C)	$1/R_1$ (units S)
A_I	current gain	$1 + R_2/R_1$
β	feedback factor	$R_1/(R_1 + R_2)$
T	loop gain	$A_d \beta$
G	closed-loop gain	$\frac{1}{\beta} \frac{T}{1+T} \rightarrow 1/\beta$
r_d, r_o	OPAMP open-loop input / output resistance	$r_d \rightarrow \infty, r_o \rightarrow 0$ (ideal)
R_{in}, R_{out}	amplifier closed-loop input / output resistance	$r_d(1+T), r_o/(1+T)$

Symbol	Meaning	Defining relation / typical
R_1, R_2	gain-setting / feedback resistors	E12 series, 5 %
R_3	bias-compensation resistor	$R_1 \parallel R_2$
R_L	load resistance	specification
R_S, R_g	source / generator output resistance	keep $R_1 \gg R_S$
I_{MAX}	maximum OPAMP output current	~ 20 mA
I_{fb}, I_{load}	output current into feedback / load	$I_{fb} < I_{load}/10$
$V_{cc}/V_{dd}, V_{ee}/V_{ss}$	positive / negative supply rails	V_{PS} : supply magnitude
V_C	common-mode input	$\frac{1}{2}(V_1 + V_2)$
A_C	common-mode gain	A_d/CMRR
CMRR	common-mode rejection ratio	A_d/A_C ; $20 \log_{10}$ in dB
PSRR (SVR)	power-supply rejection ratio	supply-side twin of CMRR
V_{OFF}	input offset voltage	$\sim$ mV
I_{BIAS}	input bias current	cancelled by R_3
I_{OFF}	input offset current	$\sim$ nA
f_{p1}, f_{p2}	first / second pole frequencies	-20 dB/dec each
f_{BW}	closed-loop bandwidth	GBWP/A_{V0}
f_T, GBWP	gain-bandwidth product	constant for compensated OPAMP
ϕ_M	phase margin	$180^\circ + \angle T$ at $ T = 0$ dB
SR	slew rate	I_s/C_c
C_c	compensation capacitor	sets SR and dominant pole
V_{OH}, V_{OL}	comparator high / low output levels	near the rails
V_{S1}, V_{S2}	upper / lower switching thresholds	$V_{S1} > V_{S2}$
ΔV_S	hysteresis width	inv. $\frac{R_1}{R_1+R_2}(V_{OH} - V_{OL})$; non-inv. $\frac{R_1}{R_2}(V_{OH} - V_{OL})$
$V_{S,avg}$	hysteresis mean $\frac{1}{2}(V_{S1} + V_{S2})$	$\neq V_{REF}$
N_A	noise amplitude on v_{in}	design rule $\Delta V_S > 2N_A$
V_{REF}	reference voltage of a comparator	sets threshold centre
ε	relative error	sum of resistor tolerances

Appendix G

Example coverage map

Every example/exercise type in `iese02_opamp_2026` and where its solving method appears here.

Example(s)	Protocol / method	Where
Non-inv $G = 3, 10000, 0.707$; inv $-3, -1, -0.707$; transres. $20 \text{ mA}..10 \text{ A}$; follower	Design ideal amplifier (P 1.1)	Ch. 1
Finite A_d , $A_V = 10$, $A_d \in [100\text{k}, 500\text{k}]$	Gain error (P 2.1)	Ch. 2
R_{in}, R_{out}	Test-source (P 2.2)	Ch. 2
R_2 lower bound ($V_{PS} = 12$, $I_{load} = 15 \text{ mA}$)	I_{MAX} sizing (P 2.3)	Ch. 2
LM741 CMR/SVR = 86 dB (CMRR, PSRR)	CMRR (P 2.4)	Ch. 2
Swing vs $R_L = 0.1/0.3/0.5 \text{ k}$	Output swing (P 2.5)	Ch. 2
Offset, $A_V = 10$, $V_{OFF} = 5 \text{ mV}$, $I_{OFF} = 60 \text{ nA}$	Offset min. (P 2.6)	§2.6
Phase margin, $ T = 0 \text{ dB}$ at f_{p2}	Stability (P 3.1)	Ch. 3
Compensation methods 1, 2.1, 2.2, 3 ($A_V = 10, 100$, follower)	Compensate (P 3.2)	§3.3
GBWP Ex1 ($A_V = 12$) / Ex2 ($A_V = 100$) / Ex ($A_V = 25$)	GBWP (P 3.3)	Ch. 3
Slew rate, LM741 full-power BW, TL082	BW vs SR (P 3.4)	Ch. 3
Audio amp ($f_L = 20 \text{ Hz}..f_H = 20 \text{ kHz}$)	Audio design (P 4.1)	Ch. 4
Adder/subtractor Ex1–3, homework	Adder/subtractor (P 4.2)	Ch. 4
Integrator (stability, wind-up)	Integrator design (P 4.3)	Ch. 4
Plain threshold comparator (inv./non-inv.); noise band $\pm N_A$	Bare comparator (P 5.1)	§5.1
Inverting Schmitt ($V_{OL} = 0, V_{OH} = 5$, $V_{REF} = 2.14 \text{ V}$); ΔV_S , $V_{S,avg}$	Inverting Schmitt (P 5.2)	§5.2
Non-inverting Schmitt thresholds	Non-inv. Schmitt (P 5.3)	§5.3
Astable, $f = 50 \text{ kHz}$; duty cycle $\neq 50 \%$	Astable (P 5.4)	Ch. 5
Triangular+square, $f = 500 \text{ Hz}$; variants, unipolar	Generator (P 5.5)	§5.5
Bench: gain, step/rise time, virtual ground, R_{in}/R_{out} , Bode, audio	Characterise op-amp (P 6.1–6.2)	Ch. 6